

DATA 266 Lab 1

Combined Team Report - Team 15

Anshika Goel | Viraat Chaudhary

Repository: <https://github.com/99GoelAnshika/data266-lab1-team15>

Team ownership statement

Team 15 consists of Anshika Goel and Viraat Chaudhary. For each of the three lab tasks, each member independently trained and evaluated a meaningfully different model configuration while preserving member-scoped code, checkpoints, logs, metrics, and failure/error evidence. Anshika built the five-block pre-layer-normalized character GPT for Task 1, the mean-pool/TextCNN/BiGRU-attention sentiment suite for Task 2, and the nine-block CycleGAN whose final selected checkpoint is epoch 125 for Task 3. Viraat built a four-block post-layer-normalized character GPT for Task 1, an RNN/LSTM/TCN sentiment suite for Task 2, and a six-block resize-convolution CycleGAN for Task 3. Cross-member conclusions in this report are descriptive comparisons of frozen evidence; where preprocessing, hardware, training definitions, or multiple architectural factors differ.

Report organization and evidence policy

- Task 1: character-level GPT-style language modeling from scratch on TinyStories.
- Task 2: Yelp Polarity binary sentiment classification with three from-scratch models per member.
- Task 3: CycleGAN unpaired Monet/photo translation with two generators and two discriminators per member.
- All metrics are reported from committed result artifacts; raw logs are treated as immutable provenance.

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1. Task 1 - GPT-style character language modeling from scratch

Task 1 trains decoder-only autoregressive character language models from scratch on TinyStories. The shared objective is next-character prediction under a causal mask, followed by quantitative evaluation and qualitative generation analysis. The member implementations deliberately differ in normalization placement, depth, width, attention-head count, activation, dropout, output-weight tying, and training batch size, allowing a practical comparison of two compact GPT-style designs without using pretrained weights or prebuilt Transformer blocks.

1.1 Shared data and evaluation protocol

Item	Protocol
Dataset	roneneldan/TinyStories
Frozen revision	f54c09fd23315a6f9c86f9dc80f725de7d8f9c64
Training sequences	100,000 fixed-length sequences
Validation sequences	10,000 fixed-length sequences
Context length	256 characters
Split boundary	Story-record groups split before sequence construction
Vocabulary boundary	Built from training text only
Generation evaluation	Greedy plus temperature sampling; saved continuations and diversity metrics

1.2 Anshika Goel - five-block pre-LN character GPT

Anshika implemented a compact pre-layer-normalized decoder-only Transformer at character granularity. Attention, causal masking, residual connections, normalization, feed-forward layers, and learned positional embeddings are implemented directly rather than delegated to a prebuilt Transformer or multi-head-attention layer. The output projection is tied to the token-embedding matrix, reducing parameters and enforcing a shared input/output representation.

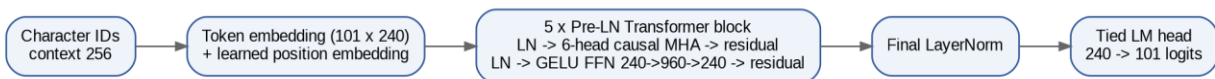


Figure 1. Anshika Task 1 architecture: five pre-LN causal Transformer blocks with tied token/output weights.

Architecture and hyperparameters

Setting	Value
Tokenization / vocabulary	Character / 101
Context length	256
Model width	240
Transformer blocks	5
Attention heads	6 (40 dimensions/head)
Feed-forward width	960
Activation	GELU
Normalization	Pre-LayerNorm
Position encoding	Learned embeddings

Setting	Value
Dropout	0.10
Output head	Tied to token embedding
Trainable parameters	3,557,861
Epochs / batch size	10 / 32
Optimizer	AdamW
Peak LR / minimum LR	3e-4 / 3e-5
Warm-up / schedule	5% / cosine decay
Weight decay	0.10

Performance and measured metrics

Metric	Value
Training cross-entropy	0.729267
Validation cross-entropy	0.693730
Validation perplexity	2.001166
Bits per character	1.000841
Generalization gap (val - train)	-0.035537
Validation top-1 next-character accuracy	78.03%
Distinct-1 / Distinct-2 / Distinct-3	0.004370 / 0.039709 / 0.158188
Repeated 4-gram rate	0.222371
Gradient norm mean / max	0.767176 / 11.639734
Loss spikes / nonfinite losses	0 / 0
AMP-skipped nonfinite-gradient updates	10
Training throughput	80,067.67 characters/s
Validation throughput	257,370.58 characters/s
Generation throughput	518.08 characters/s
Peak GPU allocated / reserved	1,586.03 / 1,754.00 MB
Total training time	55.09 minutes

Interpretation: The low validation CE and 78.03% next-character accuracy are the strongest Task 1 quality results in the member comparison. The negative saved gap uses Anshika's epoch-average training-loss convention, so it should not be numerically equated with Viraat's dropout-off full-training-set inference gap.

Character GPT Training and Validation Loss

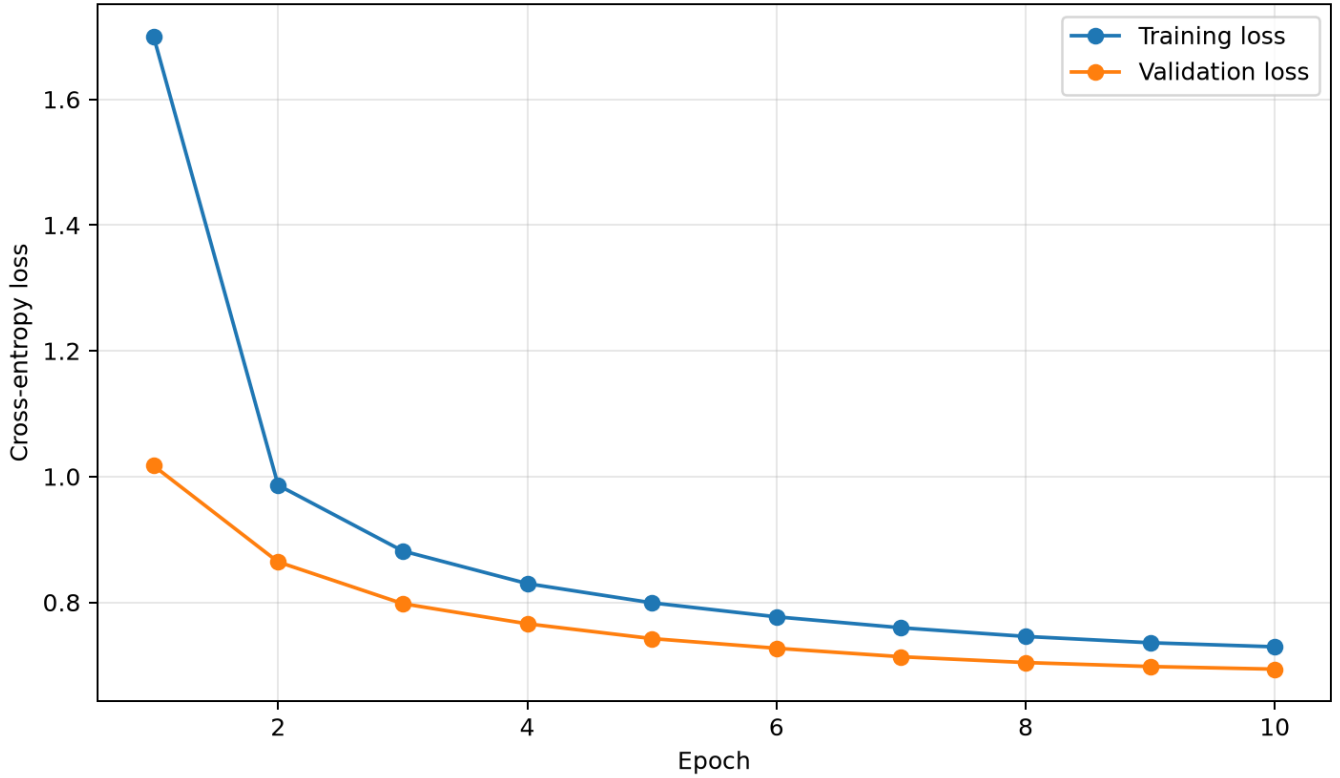


Figure 2. Anshika Task 1 training and validation loss across ten epochs.

```

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Figure 3. Final training-log evidence.

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4    "experiment_name": "anshika_char_gpt_v1",
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6    "dataset_revision": "f54c09fd23315a6f9c86f9dc80f725de7d8f9c64",
7    "source_split": "train",
8    "source_dataset_fingerprint": "9677945e7c100bfa",
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14   "row_length_including_target_shift": 257,
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16   "validation_sequences": 10000,
17   "train_characters": 25700000,
18   "validation_characters": 2570000,
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23     28609
24   ],
25   "validation_shuffled_index_range": [
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27     31468
28   ],
29   "story_overlap_between_splits": false,
30   "vocabulary_size": 101,
31   "unknown_token": "<UNK>",
32   "validation_unknown_characters": [],
33   "validation_unknown_character_count": 0,
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37   ],
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40     257
41   ],
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44   "validation_text_sha256": "6d62cdf212a2b74fb5570cd217d992d50a8adc1852b5bc99faf8858c6a73349f",
45   "train_tensor_sha256": "13a7a30403251d127baedc26f19afc7bc0d11b913534de675ea49e894ad11c7b",
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Figure 4. Split and dataset provenance.

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4    "member_name": "Anshika Goel",
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14     "path": "task1_llm/Anshika_Goel/configs/gpt_char.yaml",
15     "sha256": "31162ce40fadfea446f83570eebfb6a674a5c9ed27f6d41b136b37e6c89d287b",
16     "generation_seed": 20260925,
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19       "Lily found a little",
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25       1.3
26     ],
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29     "max_new_characters": 500
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33     "vocabulary_size": 101
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42     "gradient_norm_max": 11.639734268188477,
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183         "repeated_4gram_rate": 0.22237126462478576,
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195             "minimum_sample_repeated_4gram_rate": 0.19517102615694165,
196             "maximum_sample_repeated_4gram_rate": 0.3501006036217304
197         },
198         "1.0": {
199             "sample_count": 9,
200             "generated_tokens": 4500,
201             "distinct_1": 0.010222222222222223,
202             "distinct_2": 0.0892896904920953,
203             "distinct_3": 0.2829094154395359,
204             "repeated_4gram_rate": 0.2378716744913928,
205             "minimum_sample_repeated_4gram_rate": 0.17303822937625754,
206             "maximum_sample_repeated_4gram_rate": 0.3963782696177062
207         },
208         "1.3": {
209             "sample_count": 9,
210             "generated_tokens": 4500,
211             "distinct_1": 0.011777777777777778,
212             "distinct_2": 0.10376308171899354,
213             "distinct_3": 0.33980365908076754,
214             "repeated_4gram_rate": 0.15470601386094343,
215             "minimum_sample_repeated_4gram_rate": 0.11066398390342053,
216             "maximum_sample_repeated_4gram_rate": 0.193158953722334
217         }
218     }
219 }
220 }

```

Figure 5. Machine-readable evaluation evidence.

Strengths, weaknesses, limitations, and design rationale

- Strengths: best observed common-validation CE and next-character accuracy; tied output weights; stable ten-epoch optimization; compact 3.56M-parameter scale.
- Weaknesses: greedy generation can collapse into exact repetition; sampled text can repeat words or become grammatically incoherent at higher temperature.
- Limitations: character-level modeling lacks explicit word-level semantics; a 256-character context constrains long-range story planning; diversity metrics do not directly measure narrative coherence.
- Why this design: pre-LN is a stability-oriented Transformer layout, GELU provides a smooth nonlinear feed-forward transform, and weight tying reduces parameters while keeping input/output character representations aligned.

Failure analysis - exact observed cases

Case 1 - Sentence-level repetition loop

sample_001 | greedy | repeated 4-gram rate 0.543260

She said they were going to the park to play with it. She said they were going to the park to play with it. She said they were going to the park to play with it. She said they were going to the park to play with it.

The continuation initially remains plausible but becomes trapped in an exact sentence loop. Greedy decoding offers no randomness or explicit repetition control once a locally probable sequence dominates the recent 256-character context.

Case 2 - Pathological word repetition

sample_005 | temperature 1.0 | repeated 4-gram rate 0.396378

she found a box of blue blue blue blue blue balls. She brought it to the blue blue blue blue blue blue blanket! The blue blue blue blue blue blue, near a tree and carrots.

Repeated emission of the word blue prevents narrative progression even under temperature sampling. A future test could compare an n-gram repetition penalty or unlikelihood objective using the same prompt/seed.

Case 3 - Grammar and semantic coherence breakdown

sample_008 | temperature 1.3 | repeated 4-gram rate 0.171026

The nlight was nothing happened, hugged and ran off for bed. Moral. Joey was so happy to have opened a bowl very year of.

Malformed tokens and broken clause structure make the story difficult to interpret. A controlled temperature comparison can test whether lower sampling temperature improves grammar without increasing repetition.

1.3 Viraat Chaudhary - four-block post-LN character GPT

Viraat implemented a four-block post-layer-normalized character GPT with a wider 256-dimensional state, eight attention heads, ReLU feed-forward activation, and an untied output projection. The design uses explicit tensor-level attention, causal masking, and normalization. It therefore remains from-scratch while providing a materially different architecture from Anshika's pre-LN/GELU/tied-head model.



Figure 6. Viraat Task 1 architecture: four post-LN causal Transformer blocks with an untied language-model head.

Architecture and hyperparameters

Setting	Value
Tokenization / vocabulary	Character / 96
Context length	256
Model width	256
Transformer blocks	4
Attention heads	8
Feed-forward width	1024
Activation	ReLU
Normalization	Post-LayerNorm
Dropout	0.15

Setting	Value
Output head	Untied
Trainable parameters	3,273,824
Epochs / batch size	10 / 64
Optimizer	AdamW
Peak LR / minimum LR	3e-4 / 3e-5
Warm-up / schedule	10% / cosine decay
Precision	BF16

Performance and measured metrics

Metric	Value
Training CE, selected checkpoint FP32/dropout off	0.85758417
Final-epoch online training CE	0.93269972
Validation CE, FP32/dropout off	0.86370217
Validation perplexity	2.3719257
Bits per character	1.2460588
Generalization gap	0.0061180
Validation top-1 next-character accuracy	72.7812%
Distinct-1 / Distinct-2 / Distinct-3	0.004593 / 0.042158 / 0.166592
Repeated 4-gram rate, sampled outputs	0.174678
Gradient norm mean / max	0.624755 / 3.309997
Loss spikes / nonfinite loss / nonfinite grad	0 / 0 / 0
Training throughput	590,870.73 target characters/s
Generation throughput	696.01 characters/s
Peak GPU allocated / reserved	1,758.95 / 2,034 MiB
Total training time	451.62 s (7.53 min)
Hardware	NVIDIA A100-SXM4-40GB

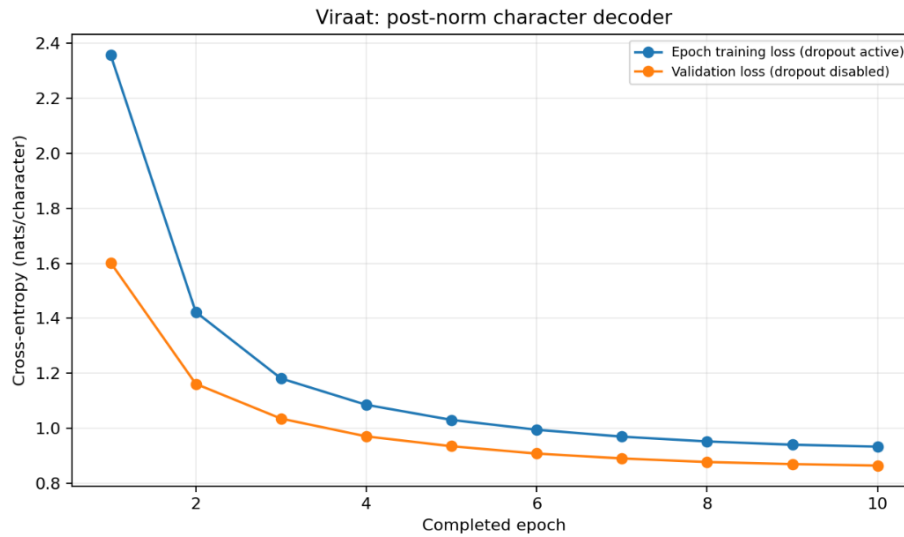


Figure 7. Viraat Task 1 training and validation loss across ten epochs.

metric	value	checkpoint_epoch	evidence
training_cross_entropy_loss	0.857584167	10	outputs/metrics/evaluation_metrics.json
final_epoch_training_cross_entropy_loss	0.932699717	10	outputs/metrics/evaluation_metrics.json
validation_cross_entropy_loss	0.863702166	10	outputs/metrics/evaluation_metrics.json
perplexity	2.371925722	10	outputs/metrics/evaluation_metrics.json
bits_per_character	1.246058832	10	outputs/metrics/evaluation_metrics.json
generalization_gap	0.006118	10	outputs/metrics/evaluation_metrics.json
top1_next_character_accuracy	0.727812109	10	outputs/metrics/evaluation_metrics.json
distinct_1	0.004592593	10	outputs/metrics/evaluation_metrics.json
distinct_2	0.042158391	10	outputs/metrics/evaluation_metrics.json
distinct_3	0.166592295	10	outputs/metrics/evaluation_metrics.json
repeated_4gram_rate	0.174677696	10	outputs/metrics/evaluation_metrics.json
gradient_norm_mean	0.624754525	10	outputs/metrics/evaluation_metrics.json
gradient_norm_max	3.309996843	10	outputs/metrics/evaluation_metrics.json
loss_spike_count	0	10	outputs/metrics/evaluation_metrics.json
nonfinite_loss_count	0	10	outputs/metrics/evaluation_metrics.json
nonfinite_gradient_count	0	10	outputs/metrics/evaluation_metrics.json
parameter_count	3273824	10	outputs/metrics/evaluation_metrics.json
training_tokens_per_second	590870.7276	10	outputs/metrics/evaluation_metrics.json
total_training_seconds	451.6223641	10	outputs/metrics/evaluation_metrics.json
peak_gpu_allocated_mb	1758.94873	10	outputs/metrics/evaluation_metrics.json
peak_gpu_reserved_mb	2034	10	outputs/metrics/evaluation_metrics.json
peak_cpu_rss_mb	1674.898438	10	outputs/metrics/evaluation_metrics.json
generation_tokens_per_second	696.0130767	10	outputs/metrics/evaluation_metrics.json

Figure 8. Viraat Task 1 full metric table.

Strengths, weaknesses, limitations, and design rationale

- Strengths: substantially faster recorded training and generation throughput; slightly smaller parameter count; lower sampled repeated-4-gram rate than Anshika.
- Weaknesses: higher common CE and lower next-character accuracy; greedy output still shows severe repetition; sampled output can switch character identity and create malformed words.
- Limitations: the post-LN/ReLU/untied design differs in several factors simultaneously, so quality differences cannot be attributed to one architectural choice; character-level context remains limited to 256 characters.
- Why this design: fewer blocks with greater width and eight heads trade depth for parallel representation capacity, while the untied head provides a separate learned output mapping. BF16 and batch 64 prioritize training efficiency.

Failure analysis - exact observed cases

Case 1 - Repetitive generation loop

sample_001 | greedy | full greedy repeated 4-gram rate 0.748491

You are very sad. You are a good friend. You are very sad. You are a good friend. You are very sad. You are a good friend.

The same sentence pair repeats without developing the story. The aggregate 0.174678 repeated-4-gram metric applies to the 27 sampled continuations and excludes greedy output, so this greedy-case rate must be interpreted separately.

Case 2 - Unexplained character-name change

sample_006 | temperature 1.0

, there was a little girl called Jelie, "Zoomp! I love you. Thank you, Grandma!" Lily said.

The continuation introduces Jelie and immediately attributes speech to Lily without establishing Lily as another character. A controlled continuation-consistency test would be needed to evaluate whether further training improves name stability.

Case 3 - Malformed words and grammar

sample_018 | temperature 1.3

Lily noticed down the tib legs. She splashed the stame. She's notice and kicked some chone flowersh.

Nonstandard forms such as stame, chone, and flowersh plus malformed grammar reduce readability. A matched-seed temperature study can test readability/repetition tradeoffs.

1.4 Task 1 team comparison

Property / metric	Anshika	Viraat
Architecture	5-block pre-LN, d=240, 6 heads, GELU, tied head	4-block post-LN, d=256, 8 heads, ReLU, untied head
Parameters	3,557,861	3,273,824
Validation CE	0.693730	0.863702
Validation top-1 accuracy	78.03%	72.7812%
Perplexity	2.001166	2.371926
Sampled repeated 4-gram rate	0.222371	0.174678
Training throughput	80,067.67 chars/s	590,870.73 chars/s
Training time	55.09 min	7.53 min
Generation throughput	518.08 chars/s	696.01 chars/s

Common held-out metric	Anshika	Viraat
Common validation CE	0.69440728	0.86098925
Common next-character accuracy	78.1023%	72.9764%

Interpretation: Under the shared held-out comparison, Anshika is the stronger quality model because it achieves lower CE and higher next-character accuracy. Viraat is the stronger efficiency model because it trains much faster, uses slightly fewer parameters, and generates faster. Because normalization, depth, width, heads, activation, dropout, weight tying, batch size, precision, and split seed differ together, the comparison supports a practical model choice rather than a causal architectural ablation.

member	common_validation_ce	common_next_character_accuracy	unknown_character_count	checkpoint_epoch
Anshika_Goel	0.69440728	0.781023047	5	10
Viraat_Choudhary	0.860989245	0.729764453	8	10

Figure 9. Common-validation comparison evidence.

Selection criterion and result

For this report, use lower cross-entropy on the common held-out text as the quality criterion, with next-character accuracy as corroborating evidence. This criterion is documented at reporting time; it is not claimed to have been preregistered before training. Both checkpoints were frozen before the common evaluation. No architecture-only causal conclusion follows from the comparison.

Preferred Task 1 checkpoint: Anshika's epoch-10 model. It has lower common CE and higher common accuracy. Retain Viraat's implementation, metrics, checkpoint and failure analysis as his separate individual contribution.

Common held-out metric	Anshika	Viraat
Cross-entropy, nats/target character	0.6944072781	0.8609892454
Next-character accuracy	78.1023%	72.9764%
Unknown encoded characters	5	8
Frozen checkpoint epoch	10	10

Anshika's accuracy is higher by 5.1259 percentage points. Her common CE is lower by 0.16658197 nats/character.

Figure 10. Team selection rationale.

1.4.1 Complete Task 1 team comparison matrix

The matrix below consolidates the architecture, hyperparameters, and every required Task 1 evaluation metric already reported in Sections 1.2-1.4. Training-CE conventions differ: Anshika reports the epoch-average training CE, while Viraat also preserves a selected-checkpoint FP32/dropout-off CE; those values are therefore shown with their original definitions rather than treated as directly interchangeable.

Property / metric	Anshika	Viraat
Architecture	5-block pre-LN decoder-only GPT; d=240; 6 heads; GELU; tied LM head	4-block post-LN decoder-only GPT; d=256; 8 heads; ReLU; untied LM head
Vocabulary / context	101 characters / 256	96 characters / 256
Feed-forward width	960	1024
Dropout	0.10	0.15
Epochs / batch	10 / 32	10 / 64
Optimizer	AdamW	AdamW
Peak LR / minimum LR	3e-4 / 3e-5	3e-4 / 3e-5
Warm-up / schedule	5% / cosine decay	10% / cosine decay
Precision	FP32 training path reported	BF16
Parameters	3,557,861	3,273,824
Training CE (reported convention)	0.729267 (epoch-average)	0.85758417 (selected checkpoint FP32/dropout off); 0.93269972 final-epoch online
Validation CE	0.693730	0.86370217
Perplexity	2.001166	2.3719257
Bits per character	1.000841	1.2460588
Generalization gap	-0.035537	0.0061180
Top-1 next-character accuracy	78.03%	72.7812%
Distinct-1 / 2 / 3	0.004370 / 0.039709 / 0.158188	0.004593 / 0.042158 / 0.166592
Repeated 4-gram rate	0.222371	0.174678
Gradient norm mean / max	0.767176 / 11.639734	0.624755 / 3.309997
Stability events	Loss spikes 0; nonfinite losses 0; AMP-skipped nonfinite-gradient updates 10	Loss spikes 0; nonfinite loss 0; nonfinite grad 0
Training throughput	80,067.67 chars/s	590,870.73 target chars/s
Generation throughput	518.08 chars/s	696.01 chars/s
Peak GPU allocated / reserved	1,586.03 / 1,754.00 MB	1,758.95 / 2,034 MiB
Total training time	55.09 min	451.62 s (7.53 min)
Common held-out CE	0.69440728	0.86098925
Common held-out next-char accuracy	78.1023%	72.9764%

1.4.2 What we would try next

- Run a controlled one-factor-at-a-time ablation on the same frozen 100,000/10,000 sequence protocol: match width, batch size, precision, learning-rate schedule and context length, then vary pre-LN versus post-LN, GELU versus ReLU, and tied versus untied output weights separately. This is needed because Anshika wins the common held-out comparison (CE 0.69440728; accuracy 78.1023%) while Viraat is much faster (590,870.73 vs 80,067.67 chars/s), but several design factors currently change together.

- Repeat the existing generation protocol with matched prompts/seeds across greedy and sampled decoding and evaluate repeated 4-gram rate together with malformed-word/name-consistency failures. Viraat has the lower sampled repeated-4-gram rate (0.174678 vs 0.222371), yet its greedy failure reaches 0.748491 on sample_001; Anshika also shows grammar/coherence failures at higher temperature, so the next experiment should quantify the quality-repetition trade-off rather than rely on isolated samples.

2. Task 2 - Yelp Polarity sentiment classification

Task 2 evaluates from-scratch neural text classifiers on Yelp Polarity. Each member trained a baseline plus two meaningfully different experimental models without pretrained embeddings or pretrained language models. Model selection was performed on validation data before official-test evaluation. The final report emphasizes not only accuracy and macro-F1 but also ranking quality, calibration, robustness slices, statistical confidence, efficiency, and a frozen 20-error qualitative review.

2.1 Shared task protocol

Item	Value
Dataset	fancyzhx/yelp_polarity
Official train / test	560,000 / 38,000
Frozen train / validation	504,000 / 56,000
Maximum sequence length	256 tokens
Training vocabulary	50,000 tokens
Decision threshold	0.50
Official-test boundary	Test accessed only after checkpoints/model choices were frozen
Required manual review	20 errors: 5 confident FP, 5 confident FN, 5 near-threshold, 5 slice-specific

2.1.1 Exploratory data analysis summary

The raw-data audit was run on the pinned Yelp Polarity revision before model training. The official train and test splits are exactly balanced, contain no missing/malformed-label rows, and show no exact stripped-text overlap across splits. Negative reviews are materially longer than positive reviews, which is relevant to the fixed 256-token input boundary used by both members.

EDA metric	Official train	Official test
Official split size	560,000	38,000
Negative / positive	280,000 / 280,000	19,000 / 19,000
Positive fraction	50.00%	50.00%
Null / non-string / blank text	0 / 0 / 0	0 / 0 / 0
Invalid labels / duplicate rows	0 / 0	0 / 0
Raw words: mean / median	136.21 / 99	135.75 / 99
Raw words: p95 / p99 / max	382 / 625 / 1046	377 / 617 / 1005
Negative mean raw words	154.92	154.77
Positive mean raw words	117.51	116.73
Processed-data metric	Recorded value	
Frozen train / validation / test	504,000 / 56,000 / 38,000	
Vocabulary / max sequence length	50,000 / 256 tokens	
Processed train length mean / median / p95 / p99	66.15 / 48 / 185 / 302	
Train truncated	8,983 (1.782%)	
Validation truncated	949 (1.695%)	
Test truncated	651 (1.713%)	
Empty after preprocessing (train / val / test)	37 / 3 / 1	
OOV fraction before truncation (train / val / test)	0.459% / 0.566% / 0.543%	
Cross-split exact-text digest overlap	0	

2.2 Anshika Goel - Mean Pool, TextCNN, and BiGRU-Attention

Anshika used a shared learned 128-dimensional word embedding and compared three representations of increasing contextual capacity. The baseline discards token order through masked mean pooling, TextCNN captures local n-gram patterns through three convolution widths, and BiGRU-Attention models bidirectional sequential context before learning a weighted document representation.

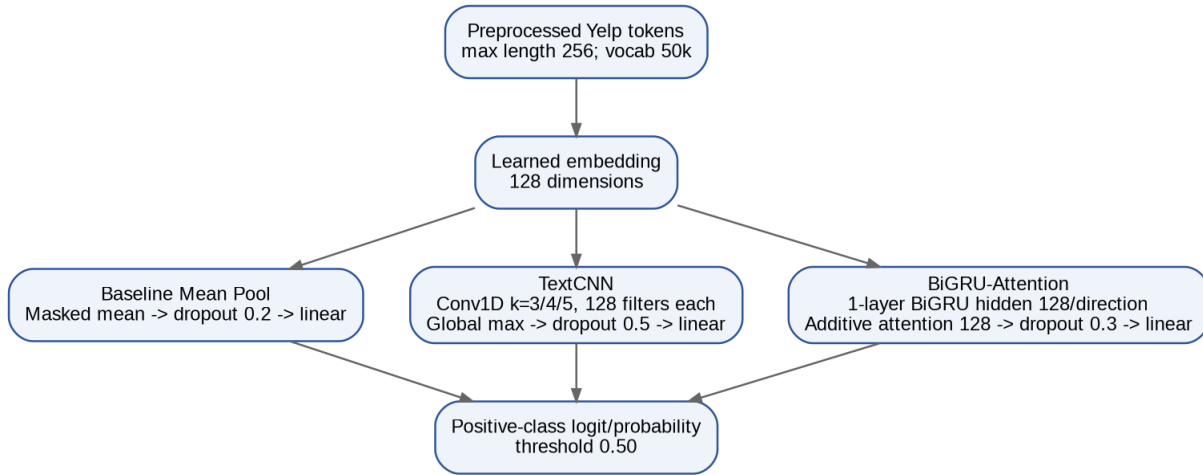


Figure 11. Anshika Task 2 model lineup: mean pooling, multikernel TextCNN, and BiGRU with additive attention.

Preprocessing and shared training controls

- Lowercase, normalize HTML/whitespace, expand negation contractions, remove punctuation and special characters, remove stopwords while preserving no/nor/not/never, apply Porter stemming, and tokenize with regex word tokens.
- Training seed 20260924; AMP enabled; gradient clip norm 1.0; 5% warm-up; minimum LR 1e-5; early-stopping patience 2; fixed decision threshold 0.50.

Model	Architecture	Dropout	Batch	Max epochs	LR	Parameters
Mean Pool	Embedding 128 -> masked mean -> linear	0.20	512	5	0.001	6,400,129
TextCNN	Embedding 128 -> Conv k=3/4/5 (128 each) -> global max -> linear	0.50	256	5	0.001	6,597,377
BiGRU-Attention	Embedding 128 -> 1-layer BiGRU h=128/dir -> additive attention 128 -> linear	0.30	128	5	0.0005	6,631,425

Final official-test classification metrics

Model	Accuracy	Macro-F1	MCC	ROC-AUC	PR-AUC	Brier	ECE
Mean Pool	0.929474	0.929474	0.858947	0.977648	0.977576	0.053777	0.008339
TextCNN	0.941737	0.941736	0.883491	0.985671	0.986079	0.044814	0.017341
BiGRU-Attn	0.944868	0.944864	0.889869	0.988415	0.988652	0.040506	0.007061

Model	Confusion matrix [[TN,FP],[FN,TP]]	Training s	Mean train ex/s	Peak GPU alloc MB	Final-test ex/s
Mean Pool	[[17661,1339],[1341,17659]]	72.678	36,636.53	269.09	37,655.44
TextCNN	[[17833,1167],[1047,17953]]	358.845	7,310.55	308.88	15,632.27
BiGRU-Attn	[[17789,1211],[884,18116]]	1545.914	1,401.12	223.45	6,407.57

Interpretation: BiGRU-Attention has the best observed accuracy, macro-F1, MCC, ROC-AUC, PR-AUC, Brier score, and ECE among Anshika's three models, but it is dramatically slower than the mean-pool baseline and TextCNN. TextCNN provides a middle ground: a sizable accuracy gain over mean pooling at much lower cost than BiGRU-Attention.

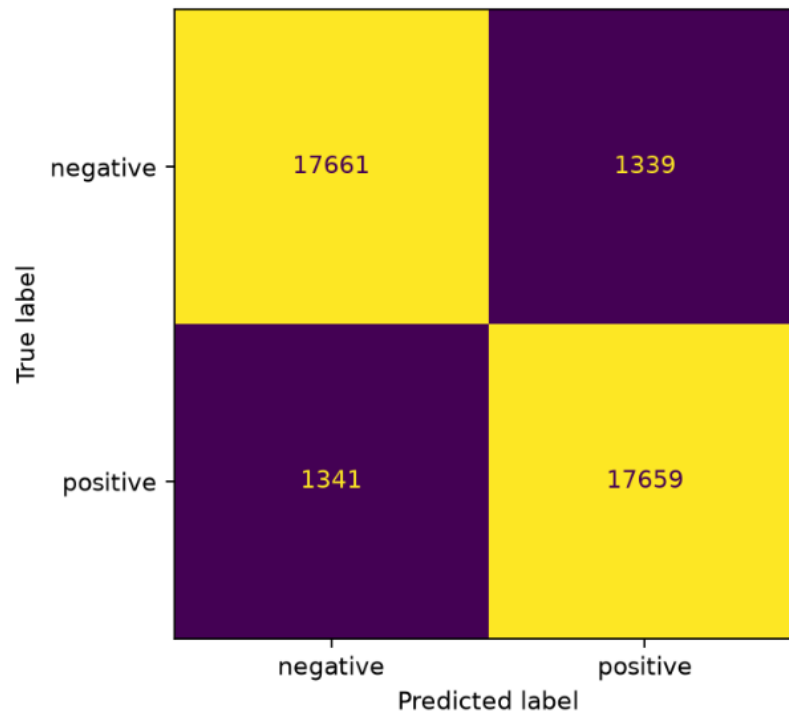


Figure 12. Anshika Mean Pool official-test confusion matrix.

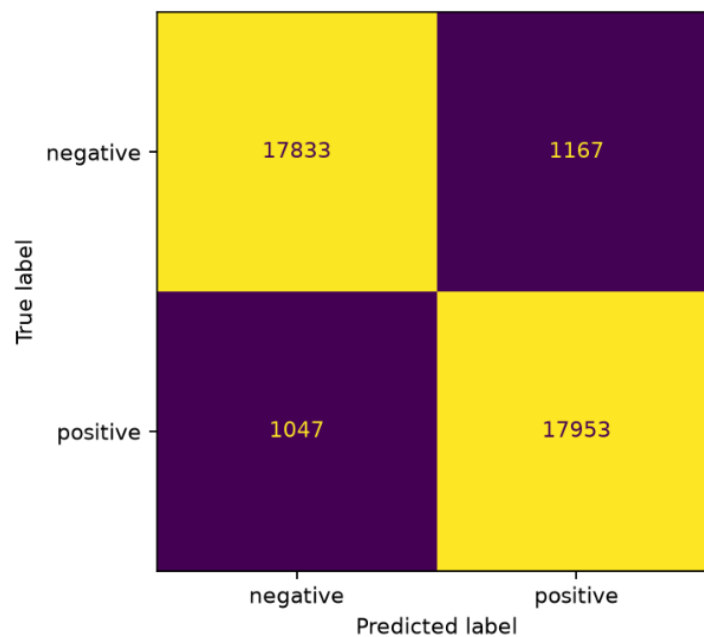


Figure 13. Anshika TextCNN official-test confusion matrix.

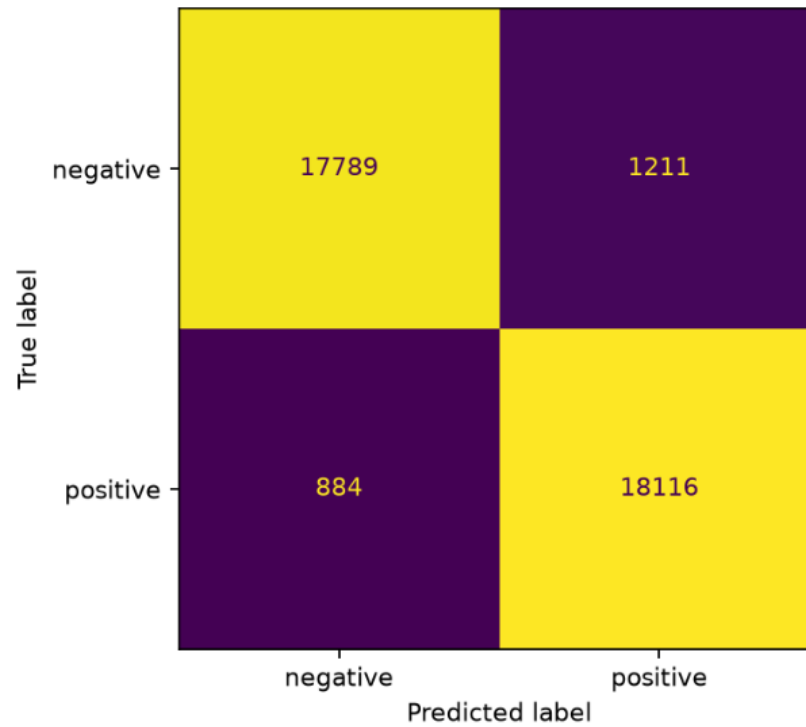


Figure 14. Anshika BiGRU-Attention official-test confusion matrix.

```

13  "models": {
14    "baseline_mean_pool": {
15      "accuracy": 0.9294736842105263,
16      "precision_macro": 0.9294736889692375,
17      "precision_micro": 0.9294736842105263,
18      "precision_weighted": 0.9294736889692378,
19      "recall_macro": 0.9294736842105262,
20      "recall_micro": 0.9294736842105263,
21      "recall_weighted": 0.9294736842105263,
22      "f1_macro": 0.9294736840151625,
23      "f1_micro": 0.9294736842105263,
24      "f1_weighted": 0.9294736840151626,
25      "mcc": 0.8589473731797639,
26      "roc_auc": 0.9776479916897507,
27      "pr_auc": 0.9775760876007973,
28      "brier_score": 0.05377662483133172,
29      "ece_15_bins": 0.008339227256021996,
30      "confusion_matrix": [
31        [
32          17661,
33          1339
34        ],
35        [
36          1341,
37          17659
38        ]
39      ],
40      "loss": 0.1925286834930119,
41      "evaluation_seconds": 1.009150299942121,
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43      "parameter_count": 6400129
44    },

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45     "textcnn": {
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47         "precision_macro": 0.9417544633359336,
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49         "precision_weighted": 0.9417544633359336,
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53         "f1_macro": 0.9417362610818281,
54         "f1_micro": 0.9417368421052632,
55         "f1_weighted": 0.9417362610818281,
56         "mcc": 0.883491305265469,
57         "roc_auc": 0.9856706343490305,
58         "pr_auc": 0.9860791532130821,
59         "brier_score": 0.04481429661252991,
60         "ece_15_bins": 0.017341390528176962,
61         "confusion_matrix": [
62             [
63                 17833,
64                 1167
65             ],
66             [
67                 1047,
68                 17953
69             ]
70         ],
71         "loss": 0.1618210131808331,
72         "evaluation_seconds": 2.430869299918413,
73         "examples_per_second": 15632.267848080268,
74         "parameter_count": 6597377
75     },

```

```

76     "bigru_attention": {
77         "accuracy": 0.9448684210526316,
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81         "recall_macro": 0.9448684210526316,
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84         "f1_macro": 0.9448643382263312,
85         "f1_micro": 0.9448684210526316,
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88         "roc_auc": 0.9884153060941829,
89         "pr_auc": 0.988652130829139,
90         "brier_score": 0.04050608959962163,
91         "ece_15_bins": 0.007061472972756902,
92         "confusion_matrix": [
93             [
94                 17789,
95                 1211
96             ],
97             [
98                 884,
99                 18116
100            ]
101        ],
102        "loss": 0.1387595862903093,
103        "evaluation_seconds": 5.930485300021246,
104        "examples_per_second": 6407.57005162189,
105        "parameter_count": 6631425
106    }
107 },

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Figure 15. Final official-test metric record.

```

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task2_sentiment > Anshika_Goel > outputs > metrics > () bootstrap_confidence_intervals.json
1 {
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3   "sampling": "with_replacement",
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5   "replicates": 2000,
6   "confidence_level": 0.95,
7   "seed": 2662502,
8   "intervals": {
9     "baseline_mean_pool": {
10      "accuracy": {
11        "lower": 0.9269730263157895,
12        "upper": 0.9319736842105263,
13        "point_estimate": 0.9294736842105263
14      },
15      "macro_f1": {
16        "lower": 0.9269699196501038,
17        "upper": 0.9319701264648411,
18        "point_estimate": 0.9294736840151625
19      },
20      "mcc": {
21        "lower": 0.8539446225551273,
22        "upper": 0.8639412680345181,
23        "point_estimate": 0.8589473731797639
24      }
25    },
26    "textcnn": {
27      "accuracy": {
28        "lower": 0.9393151315789474,
29        "upper": 0.9440526315789474,
30        "point_estimate": 0.9417368421052632
31      },
32      "macro_f1": {
33        "lower": 0.9393102254955834,
34        "upper": 0.9440521896996616,
35        "point_estimate": 0.9417362610818281
36      },
37      "mcc": {
38        "lower": 0.8786393025643553,
39        "upper": 0.8881258994357549,
40        "point_estimate": 0.883491305265469
41      }
42    },
43    "bigru_attention": {
44      "accuracy": {
45        "lower": 0.9425,
46        "upper": 0.947078947368421,
47        "point_estimate": 0.9448684210526316
48      },
49      "macro_f1": {
50        "lower": 0.9424986682861791,
51        "upper": 0.9470731167508879,
52        "point_estimate": 0.9448643382263312
53      },
54      "mcc": {
55        "lower": 0.8851473731035683,
56        "upper": 0.8942835168323436,
57        "point_estimate": 0.8898686423978138
58      }
59    }
60  }
61 }
62

```

```

({} mcnemar_tests.json X
task2_sentiment > Anshika_Goel > outputs > metrics > {} mcnemar_tests.json > ...
1  {
2    "test": "mcnemar",
3    "continuity_correction": true,
4    "multiple_testing": {
5      "method": "holm",
6      "family_size": 2,
7      "alpha": 0.05
8    },
9    "comparisons": {
10     "baseline_mean_pool_vs_textcnn": {
11       "a_correct_b_wrong": 805,
12       "a_wrong_b_correct": 1271,
13       "discordant_pairs": 2076,
14       "chi_square_statistic": 104.15462427745665,
15       "raw_p_value": 1.871274172580442e-24,
16       "continuity_correction": true,
17       "model_a": "baseline_mean_pool",
18       "model_b": "textcnn",
19       "holm_adjusted_p_value": 1.871274172580442e-24,
20       "reject_after_holm_at_0_05": true
21     },
22     "baseline_mean_pool_vs_bigru_attention": {
23       "a_correct_b_wrong": 648,
24       "a_wrong_b_correct": 1233,
25       "discordant_pairs": 1881,
26       "chi_square_statistic": 181.3163211057948,
27       "raw_p_value": 2.5004586339640436e-41,
28       "continuity_correction": true,
29       "model_a": "baseline_mean_pool",
30       "model_b": "bigru_attention",
31       "holm_adjusted_p_value": 5.000917267928087e-41,
32       "reject_after_holm_at_0_05": true
33     }
34   }
35 }
36

```

Figure 16. Bootstrap and paired-test evidence

Strengths, weaknesses, limitations, and design rationale

- Mean Pool is fastest and easiest to interpret, but it discards word order and interaction structure.
- TextCNN efficiently captures short local phrases and improves accuracy substantially, but global max pooling can overemphasize isolated phrases and miss long discourse reversals.
- BiGRU-Attention has the strongest observed classification and calibration metrics, but recurrent computation makes it by far the slowest model in Anshika's suite.
- The 256-token limit is a concrete representation limitation for long reviews; preprocessing can also remove punctuation/discourse cues relevant to sarcasm and mixed sentiment.
- The three-model progression was chosen to compare order-independent lexical aggregation, local n-gram modeling, and full bidirectional sequence modeling under the same dataset/evaluation protocol.

Failure/error analysis - 20 frozen official-test errors

The focal model is BiGRU-Attention. The review IDs were frozen before raw review text was accessed. The 20 cases are exactly balanced across four required categories. The following compact table preserves each frozen case identity and a verbatim excerpt from the committed failure-analysis table; ellipses already present in the source artifact remain ellipses. Full raw review text is preserved separately in the snapshot artifact.

#	Category	Row	True -> Pred	P+	Error type	Verbatim source excerpt
1	confident false positive	29330	negative -> positive	1.000000	Possible label-text disagreement	Wow love the place and everything is very clean and new! Great place to come and relax worth a try! Cheers, Eric Van Nguyen Visited April 2012
2	confident false positive	14825	negative -> positive	0.998535	Sarcasm / pragmatic polarity	Do you believe in Yin and Yang? The ancient Asian philosophy suggesting polar opposites are interrelated? If you don't, then you best start believi...
3	confident false positive	21331	negative -> positive	0.998047	Sentiment scope / entity confusion	This place is only awesome if you are one of three things: 1. an Ohio State fan, 2. a metrosexual, or 3. someone trying to hook up with an Ohio Sta...
4	confident false positive	5752	negative -> positive	0.997559	Temporal update / review-history conflict	NOTE: This was a 4-star review, but the food quality and ESPECIALLY customer service have gone down the tubes. See update below. Complaining I can't...
5	confident false positive	8426	negative -> positive	0.997559	Mixed-aspect sentiment	Saturday / Sunday AYCE brunch In true las vegas fashion, you get a flat rate to eat your heart out. For Strip food, main menu items seem reasonably...
6	confident false negative	22807	positive -> negative	0.000229	Temporal update / possible label-text disagreement	EDIT: They really did change the service up since I last posted this. Horrible service. Used to be my favorite pizza in the city (at a reasonable p...
7	confident false negative	30793	positive -> negative	0.000782	Temporal reversal / contrast	This place is so much better since they changed owners. My wife and I went when it was the old owners, it was terrible. We waited forever and the f...
8	confident false negative	30958	positive -> negative	0.001561	Long complaint with positive final resolution	Look, we all know Cox sucks. In fact they are a terrible business and their practices are laughable. However they are a necessary evil if you want ...
9	confident false negative	22451	positive -> negative	0.001782	Aspect conflict / overall-rating mismatch	The food is crap. I'm not trying to be mean, but it really is horrible. I'd rather eat one of those Tornado things from Circle K for dinner. Also, ...
10	confident false negative	25329	positive -> negative	0.002132	Compositional stance with negative lexical density	Im with Kimberly B...this is a card for people rebuilding credit. Mine got tanked due to a divorce. I have been rebuilding, and started with a \$300...

#	Category	Row	True -> Pred	P+	Error type	Verbatim source excerpt
11	near threshold error	12415	negative -> positive	0.500000	Mixed sentiment / threshold uncertainty	My friends and I were in the mood for some cheap sushi..and since I was with 5 guy friends, I was not surprised they chose an all-you-can-eat place...
12	near threshold error	16620	negative -> positive	0.500000	Informal ambivalence / lexical variation	blush is a cool place to hang out.. idk if it's a lounge. felt like it.. but with a club crowd? there's not enough places to chill and a million pp...
13	near threshold error	31278	negative -> positive	0.500000	Short-context pragmatic criticism	How hard is it to get a 2 donut order right?? One chocolate glazed and one strawberry glazed donut. That's IT. Apparently pretty hard.
14	near threshold error	37323	negative -> positive	0.500000	Mild mixed sentiment / aspect balance	They have a good space but I don't think their using it wisely... The food is okay for a bar but my martini was very good. Adding a bit more seatin...
15	near threshold error	18639	positive -> negative	0.499756	Extremely short positive context	You really need a whole day to see this centre. The gardens are spectacular, as are the views.
16	slice specific error	12480	negative -> positive	0.997070	Short-review ambiguity / possible label-text disagreement	my husband had an omelette that was good. i had a blt, a little on the small side for \$10, but bacon was great. Our server was awesome!
17	slice specific error	20203	negative -> positive	0.997559	Contrast with misleading positive cues	Not all that. This joint has some of the best reviews short of Joel Robuchon but the fact is it's not much more than a glorified protein-buffet. Th...
18	slice specific error	6611	positive -> negative	0.003111	Negative lexical density with positive final stance	Harlow's gives me diarrhea. Every time. Every single time. You know what, though? I don't care because Harlow's cuts my hangover directly in half a...
19	slice specific error	37771	negative -> positive	0.997559	Negation / location-specific reversal	I have to say I love cafe rio! Their food is simple yet delicious, the service is great as well as consistent, and my appetite and tummy are always...
20	slice specific error	26587	negative -> positive	0.990723	Sequence truncation with late decisive evidence	On the whole, this restaurant does give a new patron a great slice of European cuisine and it is worth noting that if you've never been here before...

Representative exact text and interpretation

Wow love the place and everything is very clean and new! Great place to come and relax worth a try! Cheers, Eric Van Nguyen Visited April 2012

Confident false positive, source row 29330, $P(\text{positive})=1.000000$. The visible text is strongly positive despite the frozen negative label, so this is documented as a possible label-text disagreement rather than automatically re-labeled.

This place is so much better since they changed owners. My wife and I went when it was the old owners, it was terrible. We waited forever and the food never came before we walked out...

Confident false negative, source row 30793. The review begins with positive current sentiment but contains a strongly negative historical narrative, illustrating temporal reversal and discourse weighting.

How hard is it to get a 2 donut order right?? One chocolate glazed and one strawberry glazed donut. That's IT. Apparently pretty hard.

Near-threshold negative review, source row 31278, where very short context and pragmatic criticism provide a difficult signal around the fixed 0.50 threshold.

Harlow's gives me diarrhea. Every time. Every single time. You know what, though? I don't care because Harlow's cuts my hangover directly in half and I feel twice as good when I leave...

Slice-specific false negative, source row 6611. Dense negative lexical content coexists with an ultimately positive stance, a recurring failure pattern in the frozen review.

2.3 Viraat Chaudhary - RNN, LSTM, and residual causal TCN

Viraat compared three sequence-model families with a common 128-dimensional learned embedding and masked mean-plus-maximum pooling. The baseline is a vanilla RNN, experiment 1 is an LSTM, and experiment 2 is a residual causal temporal convolutional network with exponentially increasing dilations. This contrasts recurrent memory gating against a parallel convolutional receptive field while holding many training settings fixed.

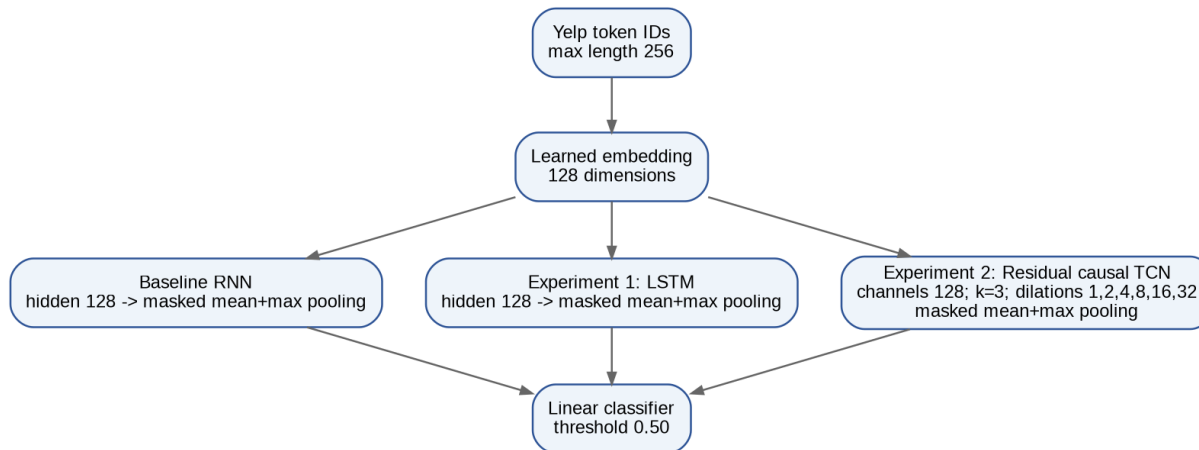


Figure 17. Viraat Task 2 lineup: RNN baseline, LSTM, and residual causal TCN.

Hyperparameters

Setting	RNN	LSTM	TCN
Embedding / hidden	128 / 128	128 / 128	128 / 128 channels
Embedding dropout / dropout	0.10 / 0.25	0.10 / 0.25	0.10 / 0.25
Batch / max epochs	512 / 5	512 / 5	512 / 5
Learning rate / weight decay	0.001 / 0.0001	0.001 / 0.0001	0.001 / 0.0001
Gradient clip / patience	1.0 / 2	1.0 / 2	1.0 / 2

Setting	RNN	LSTM	TCN
Special architecture	Vanilla RNN	LSTM gates	k=3; dilations 1,2,4,8,16,32
Parameters	6,450,049	6,549,121	7,027,969

Final official-test performance

Model	Accuracy	Macro-F1	MCC	ROC-AUC	PR-AUC	Brier	ECE
RNN	0.942658	0.942657	0.885343	0.986512	0.986900	0.042879	0.002350
LSTM	0.948474	0.948472	0.897019	0.989121	0.989355	0.038735	0.008854
TCN	0.948026	0.948021	0.896242	0.989122	0.989492	0.039448	0.011682

Model	TN	FP	FN	TP	Train s	Train ex/s	Peak CUDA MiB
RNN	17836	1164	1015	17985	46.984	58,008	581.15
LSTM	18141	859	1099	17901	57.146	48,179	712.03
TCN	18208	792	1183	17817	280.528	9,414	2,738.23

Interpretation: LSTM has the highest observed accuracy, macro-F1, and MCC and offers the strongest accuracy/time balance in Viraat's suite. TCN is nearly tied in accuracy and has marginally higher ROC-AUC/average precision plus the best long-review macro-F1, but it costs roughly 4.9x the LSTM training time and much more GPU memory. RNN has the lowest ECE and the highest speed.

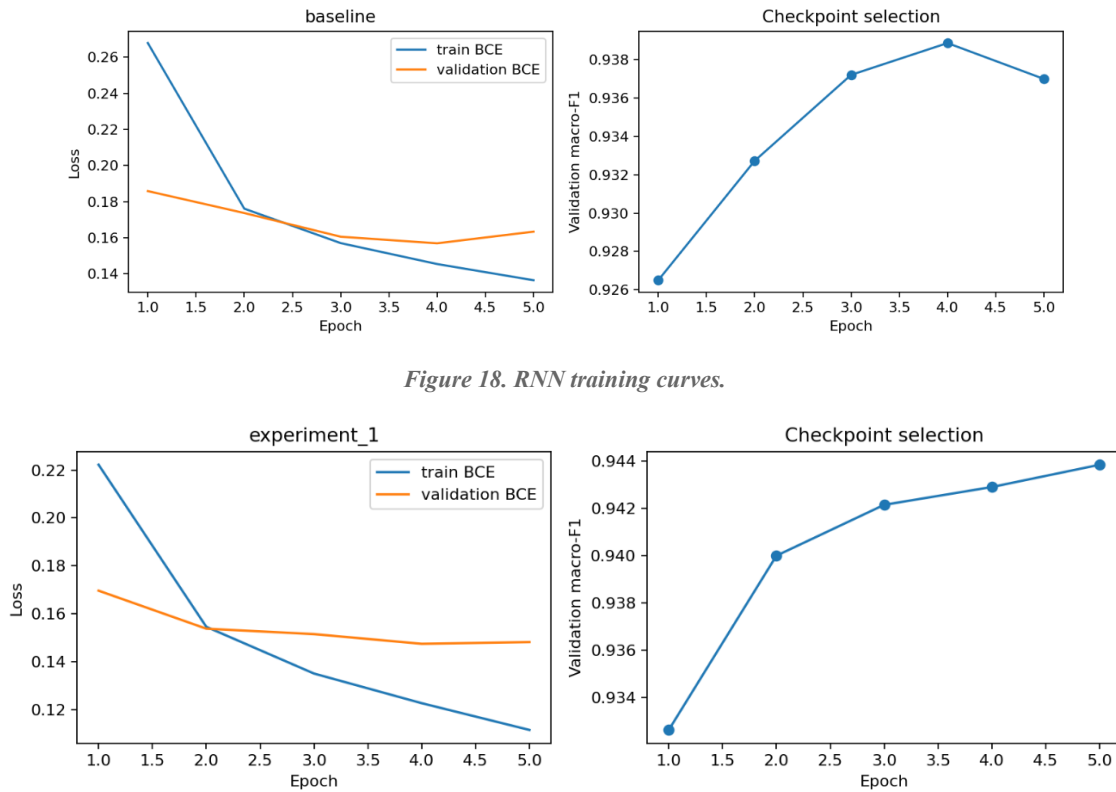


Figure 19. LSTM training curves.

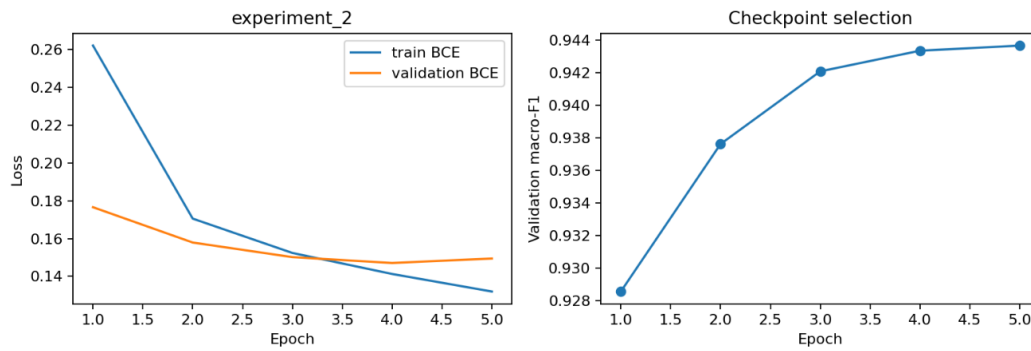


Figure 20. TCN training curves.

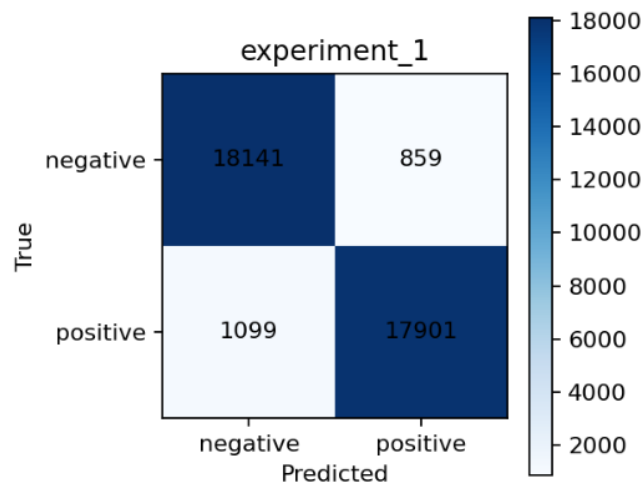


Figure 21. LSTM confusion matrix.

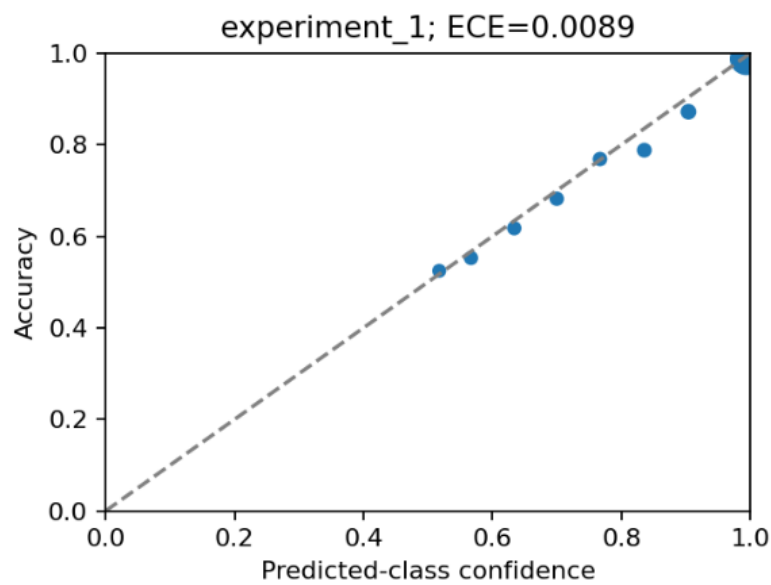


Figure 22. Viraat LSTM reliability diagram.

```

({) checkpoint_result_mapping.json X
task2_sentiment > Viraat_Chaudhary > outputs > metrics > ({) checkpoint_result_mapping.json > ...
1 {
2   "frozen_models_sha256": "c32d4ae0abe4a5f47c0b45557f9a79891af3e03be02a9f106d0a2d5d3efdea04",
3   "models": {
4     "baseline": {
5       "checkpoint": "checkpoints/baseline_best.pt",
6       "checkpoint_sha256": "fa3ae7b0eda530081858a2d471335fef949295bfea924e75c718b4738860ab48",
7       "config": "configs/baseline.json",
8       "config_sha256": "11ba31a77bea8119b90c54de3e6105b1b886b07ec1252256719b988e63a70b69",
9       "validation_manifest": "outputs/metrics/baseline_training_manifest.json",
10      "test_metrics": "outputs/metrics/final_test_metrics.json#baseline",
11      "predictions": "outputs/predictions/baseline_official_test.npz"
12    },
13    "experiment_1": {
14      "checkpoint": "checkpoints/experiment_1_best.pt",
15      "checkpoint_sha256": "ae271d65898e51dcc8aa4c3fec3c761800eda3931ef09443563a160ab5ec87bd",
16      "config": "configs/experiment_1.json",
17      "config_sha256": "bb0d7f34f7b1de96771e571b1a77c4af0eb30ea3fc3892c4ebf4d755cf9ea71c",
18      "validation_manifest": "outputs/metrics/experiment_1_training_manifest.json",
19      "test_metrics": "outputs/metrics/final_test_metrics.json#experiment_1",
20      "predictions": "outputs/predictions/experiment_1_official_test.npz"
21    },
22    "experiment_2": {
23      "checkpoint": "checkpoints/experiment_2_best.pt",
24      "checkpoint_sha256": "4bdac5813be95e42904cf2f733e2c0675528dd7ef9ef96596d8e0f5fd3d83eb6",
25      "config": "configs/experiment_2.json",
26      "config_sha256": "30507082a8277fd5648ab005fcfd9e0823d3a09d482007b49b71326795fa4a2",
27      "validation_manifest": "outputs/metrics/experiment_2_training_manifest.json",
28      "test_metrics": "outputs/metrics/final_test_metrics.json#experiment_2",
29      "predictions": "outputs/predictions/experiment_2_official_test.npz"
30    }
31  }
32 }
33

```

Figure 23. Checkpoint-to-result provenance.

```

({) experiment_1_training.json X
task2_sentiment > Viraat_Chaudhary > logs > ({) experiment_1_training.json
1 {"time_utc": "2026-10-02T01:18:28.510255+00:00", "event": "training_session_started", "model": "experiment_1", "identity": "b7a45c61c0ee7b56196a3128325411aa2e645c4de2d9fd8bc599c3c5ba771de", "config": {"
2 {"time_utc": "2026-10-02T01:18:41.326157+00:00", "event": "epoch_completed", "epoch": 1, "train_loss": 0.2222128136422899, "validation_loss": 0.16965821075439452, "validation_macro_f1": 0.932612026762214
3 {"time_utc": "2026-10-02T01:18:51.825925+00:00", "event": "epoch_completed", "epoch": 2, "train_loss": 0.15457221358257628, "validation_loss": 0.153795362779807235, "validation_macro_f1": 0.93999943397425
4 {"time_utc": "2026-10-02T01:19:04.109127+00:00", "event": "epoch_completed", "epoch": 3, "train_loss": 0.13507490754505944, "validation_loss": 0.1515317802429199, "validation_macro_f1": 0.942154615711039
5 {"time_utc": "2026-10-02T01:19:14.881839+00:00", "event": "epoch_completed", "epoch": 4, "train_loss": 0.12270394704076978, "validation_loss": 0.1474898809681353, "validation_macro_f1": 0.942910130989638
6 {"time_utc": "2026-10-02T01:19:25.718524+00:00", "event": "epoch_completed", "epoch": 5, "train_loss": 0.11153817027050351, "validation_loss": 0.14821149958883012, "validation_macro_f1": 0.94385367668106
7 {"time_utc": "2026-10-02T01:19:27.207000+00:00", "event": "training_completed", "summary": {"status": "completed", "model": "experiment_1", "identity": "b7a45c61c0ee7b56196a3128325411aa2e645c4de2d9fd8bc
8

```

Figure 24. LSTM raw training log.

Strengths, weaknesses, limitations, and design rationale

- RNN provides a fast recurrent baseline with strong calibration but lower classification quality than LSTM/TCN.
- LSTM gates improve long-range sequential modeling with only a modest runtime increase over RNN, producing the strongest overall observed Task 2 score.
- TCN offers parallel dilated context and the strongest long-review slice result, but its compute/memory cost is much higher in this implementation.
- All three are single-seed, five-epoch runs; parameter counts differ and no LSTM-vs-TCN significance test was reported, so the small 0.0447-point accuracy gap does not prove superiority.
- The same 256-token limit, preprocessing choices, high-OOV slice difficulty, and mixed-sentiment/discourse-reversal issues remain important limitations.

Failure/error analysis - 20 frozen LSTM errors

Viraat selected the LSTM using validation macro-F1 before official-test access. The committed failure analysis contains 20 unique official-test errors, five in each required group. The compact table below retains case identity, prediction metadata, error type, and an exact leading excerpt of the saved review; the complete review text and full explanation remain in the cited failure-analysis artifact.

#	Category	Row	True -> Pred	P+	Error type	Verbatim leading excerpt
1	confident false positive	29330	0 -> 1	0.999996	Possible label-text mismatch	Wow love the place and everything is very clean and new! Great place to come and relax worth a try! Cheers, Eric Van Nguyen Visited April 2012
2	confident false positive	12480	0 -> 1	0.999910	Mixed aspects with a value complaint	my husband had an omelette that was good. i had a blt, a little on the small side for \$10, but bacon was great. Our server was awesome!
3	confident false positive	34454	0 -> 1	0.999690	Contrast and authenticity criticism	It's a really good concept but the execution in authentic flavors for each region falls a little short. Every dish we tried was a bit off. We had t...
4	confident false positive	20008	0 -> 1	0.999381	Mixed sentiment and audience suitability	Fun place but when you're in your early-mid 30s and have a tad bit of style other than local canny then you will stick out like sore thumb. We're r...
5	confident false positive	27375	0 -> 1	0.999366	Unmet expectations despite positive aspects	We had dinner here for four. Had an artichoke appetizer with pita chips, club sandwich and shared a huge sundae for dessert. The service was awesom...
6	confident false negative	36001	1 -> 0	0.000089	Mixed aspects and return intention	Good food, decent beer, terrible bar service. Girl with short blond hair behind bar was awful. I'm sure this places gets better with time. I'd come...
7	confident false negative	22807	1 -> 0	0.000099	Updated opinion with historical complaint	EDIT: They really did change the service up since I last posted this. Horrible service. Used to be my favorite pizza in the city (at a reasonable p...
8	confident false negative	3223	1 -> 0	0.000102	Conflicting visits and willingness to return	I've been here two times. The first time was really good, but the second was very boring flavored. I had some lemon chicken that was mediocre. For ...
9	confident false negative	30793	1 -> 0	0.000191	Temporal reversal after ownership change	This place is so much better since they changed owners. My wife and I went when it was the old owners, it was terrible. We waited forever and the f...
10	confident false negative	3485	1 -> 0	0.000229	Resolved complaint in a final update	Update: Emailed 2 weeks ago, did not get a reply or call. . Did receive a message from Chris H and asked me to resend my email which I did today. W...

#	Category	Row	True -> Pred	P+	Error type	Verbatim leading excerpt
11	near threshold error	29664	0 -> 1	0.500223	Little sentiment evidence	I'm looking for comedians to start a comedy night in Glendale if your interested email me cutty20s@yahoo.com
12	near threshold error	15540	1 -> 0	0.499713	Positive review near the threshold	First impressions are the most important. I am so glad I stopped by this location. The lady that runs the front counter is so freaking funny. I did...
13	near threshold error	18014	1 -> 0	0.499658	Mixed rating and double negation	Honestly the quality of food is about a 3 star but because I support the mom and pop shops and used to go here almost every day during my breaks fr...
14	near threshold error	21051	1 -> 0	0.499618	Positive evaluation with incidental negation	I usually don't write reviews but reading the ones on this site helped me decide where to go, so I'm paying it forward. I haven't been to a traditi...
15	near threshold error	19979	0 -> 1	0.500658	Positive food but negative safety and management	My wife thought of a great idea to grab some carb legs for my birthday so she found Finz. Upon arriving we knew it was an old Vinnie's which at tim...
16	slice specific failure	37451	0 -> 1	0.998225	Price-value criticism in the high-OOV slice	Decor and atmosphere really good. Actually a bit upscale for its Eastowne Mall location. Food was good, but not inexpensive. See photo. Ordered rib...
17	slice specific failure	30086	1 -> 0	0.000364	Business closure with ambiguous evaluative target	Went there yesterday and found that the place has just CLOSED. Another victim of a good chef with a bad location and business planning. Hint for fu...
18	slice specific failure	29494	1 -> 0	0.001016	Long review with a positive resolution	UPDATED. My initial very frustrated and dramatic review read as follows: Bililing practices are at best negligent and at worst fraudulent. I started...
19	slice specific failure	21215	1 -> 0	0.000612	Sentiment directed at customers rather than staff	Last night several parents came in with over 15 children to celebrate their 9 year olds 4th grade graduation at 9:15 pm. The bartender expressed th...
20	slice specific failure	34919	1 -> 0	0.000856	Qualified praise after criticism	These burgers taste like they're made with high end Big Mac patties! Not a lot of variety, and really nothing all that special, but what they do, t...

Representative case 1 - Possible label-text mismatch

Wow love the place and everything is very clean and new! Great place to come and relax worth a try! Cheers, Eric Van Nguyen Visited April 2012

The text is entirely positive: it praises the clean venue and recommends visiting. The prediction agrees with the visible text but conflicts with the recorded negative label. This is a possible label-text mismatch, not proof that the official label is wrong.

Representative case 2 - Mixed aspects with a value complaint

my husband had an omelette that was good. i had a blt, a little on the small side for \$10, but bacon was great. Our server was awesome!

The food and server receive praise, but the BLT is described as small for its price. The positive prediction may reflect stronger positive wording than the value complaint. The binary rating label does not reveal how the reviewer weighted these aspects.

Representative case 18 - Long review with a positive resolution

UPDATED. My initial very frustrated and dramatic review read as follows: Bililng practices are at best negligent and at worst fraudulent. I started going to the studio per a groupon. I enjoyed the experience so much that I purchased a discounted 20 pack of classes. When I purchased the classes, my credit card was charged. However, the purchase didn't show up in my online account so I couldn't schedule classes. I called The Studio and they told me that they'd credited the classes to another student but they did remedy the situation. Fast forward about 6 months and I am reviewing my credit card statement. Lo and behold a mysterious charge from the Studio shows up for \$130. I have not even been back there in months and certainly have not authorized any transactions to this business. I am now disputing the charge via my credit card. The yoga is great but all these terrible billing practices ...

An earlier section alleges billing problems, but a later update says the owner resolved the issue beyond expectations and explicitly updates the rating to four stars. The negative prediction misses that current resolution. Long-review membership does not establish that the positive update was truncated.

Representative case 20 - Qualified praise after criticism

These burgers taste like they're made with high end Big Mac patties! Not a lot of variety, and really nothing all that special, but what they do, they do well.

The review criticizes limited variety and ordinary burgers, but concludes that the restaurant does what it offers well. The negative prediction does not match the positive label. The closing concession carries praise despite negative wording earlier in this short review.

2.4 Task 2 team comparison

Member	Model	Accuracy	Macro-F1	MCC	ROC-AUC	PR-AUC	Brier	ECE
Anshika	Mean Pool	0.929474	0.929474	0.858947	0.977648	0.977576	0.053777	0.008339
Anshika	TextCNN	0.941737	0.941736	0.883491	0.985671	0.986079	0.044814	0.017341
Anshika	BiGRU-Attn	0.944868	0.944864	0.889869	0.988415	0.988652	0.040506	0.007061
Viraat	RNN	0.942658	0.942657	0.885343	0.986512	0.986900	0.042879	0.002350
Viraat	LSTM	0.948474	0.948472	0.897019	0.989121	0.989355	0.038735	0.008854
Viraat	TCN	0.948026	0.948021	0.896242	0.989122	0.989492	0.039448	0.011682

Interpretation: Descriptively, Viraat's LSTM is the strongest overall model on accuracy, macro-F1, and MCC; the TCN is almost tied and slightly leads on ROC-AUC/average precision. Anshika's BiGRU-Attention is the strongest of her three and has lower ECE than Viraat's LSTM/TCN. Cross-member architecture effects should not be over-interpreted because preprocessing, vocabulary, hardware, training settings, and slice memberships are member-specific, and cross-member paired significance was not established.

2.4.1 Complete Task 2 team comparison matrix

The following tables consolidate the six from-scratch models using the frozen official-test predictions. Precision, recall and F1 are reported under macro, micro and weighted averaging; confidence intervals use 2,000 nonparametric bootstrap replicates; McNemar tests compare each member baseline with that member's two experiments; robustness slices retain each member's training-derived tokenizer-specific slice definitions.

Property	A: Mean Pool	A: TextCNN	A: BiGRU-Attn	V: RNN	V: LSTM	V: TCN
Architecture	Embedding 128 -> masked mean -> linear	Embedding 128 -> Conv k=3/4/5 (128 each) -> global max -> linear	Embedding 128 -> 1-layer BiGRU h=128/dir -> additive attention 128 -> linear	Embedding 128 -> vanilla RNN h=128 -> masked mean+max pooling -> classifier	Embedding 128 -> LSTM h=128 -> masked mean+max pooling -> classifier	Embedding 128 -> residual causal TCN, k=3, dilations 1/2/4/8/16/32 -> classifier
Dropout	0.20	0.50	0.30	embed 0.10 / model 0.25	embed 0.10 / model 0.25	embed 0.10 / model 0.25
Batch size	512	256	128	512	512	512
Max epochs	5	5	5	5	5	5
Learning rate	0.001	0.001	0.0005	0.001	0.001	0.001
Weight decay	reported shared control	reported shared control	reported shared control	0.0001	0.0001	0.0001
Gradient clip	1.0	1.0	1.0	1.0	1.0	1.0
Parameters	6,400,129	6,597,377	6,631,425	6,450,049	6,549,121	7,027,969
Property	A: Mean Pool	A: TextCNN	A: BiGRU-Attn	V: RNN	V: LSTM	V: TCN
Accuracy	0.929474	0.941737	0.944868	0.942658	0.948474	0.948026
Precision macro	0.929474	0.941754	0.945000	0.942685	0.948545	0.948216
Recall macro	0.929474	0.941737	0.944868	0.942658	0.948474	0.948026
F1 macro	0.929474	0.941736	0.944864	0.942657	0.948472	0.948021
Precision micro	0.929474	0.941737	0.944868	0.942658	0.948474	0.948026
Recall micro	0.929474	0.941737	0.944868	0.942658	0.948474	0.948026
F1 micro	0.929474	0.941737	0.944868	0.942658	0.948474	0.948026
Precision weighted	0.929474	0.941754	0.945000	0.942685	0.948545	0.948216
Recall weighted	0.929474	0.941737	0.944868	0.942658	0.948474	0.948026
F1 weighted	0.929474	0.941736	0.944864	0.942657	0.948472	0.948021
MCC	0.858947	0.883491	0.889869	0.885343	0.897019	0.896242
ROC-AUC	0.977648	0.985671	0.988415	0.986512	0.989121	0.989122
PR-AUC	0.977576	0.986079	0.988652	0.986900	0.989355	0.989492
Brier score	0.053777	0.044814	0.040506	0.042879	0.038735	0.039448
ECE (15 bins)	0.008339	0.017341	0.007061	0.002350	0.008854	0.011682
Confusion TN/FP/FN/TP	17661/1339/1341/17659	17833/1167/1047/17953	17789/1211/884/1116	17836/1164/1015/17985	18141/859/1099/17901	18208/792/1183/17817
Property	A: Mean Pool	A: TextCNN	A: BiGRU-Attn	V: RNN	V: LSTM	V: TCN
Training time (s)	72.678	358.845	1545.914	46.984	57.146	280.528
Train examples/s	36636.53	7310.55	1401.12	58008.29	48179.36	9414.18
Peak GPU allocated (MiB)	269.09	308.88	223.45	581.15	712.03	2738.23
Final-test examples/s	37655.44	15632.27	6407.57	22669.21	178045.15	19204.01
Member / scope		Exact CPU		Exact GPU		
Anshika - all three models		12th Gen Intel(R) Core(TM) i7-12650H (10 physical / 16 logical cores)		NVIDIA GeForce RTX 3060 Laptop GPU (6144 MiB)		
Viraat - all three models		Intel(R) Xeon(R) CPU @ 2.20GHz		NVIDIA A100-SXM4-40GB (39.49 GiB)		
Model	95% CI accuracy		95% CI macro-F1		95% CI MCC	
Anshika Mean Pool	[0.926973, 0.931974]		[0.926970, 0.931970]		[0.853945, 0.863941]	
Anshika TextCNN	[0.939315, 0.944053]		[0.939310, 0.944052]		[0.878639, 0.888126]	
Anshika BiGRU-Attn	[0.942500, 0.947079]		[0.942499, 0.947073]		[0.885147, 0.894284]	
Viraat RNN	[0.940211, 0.944947]		[0.940209, 0.944947]		[0.880427, 0.889931]	
Viraat LSTM	[0.946263, 0.950606]		[0.946252, 0.950602]		[0.892601, 0.901297]	
Viraat TCN	[0.945841, 0.950184]		[0.945826, 0.950177]		[0.891877, 0.900556]	
McNemar comparison	A correct/B wrong	A wrong/B correct	Discordant	Chi-square	Holm p	Reject 0.05
Anshika Mean Pool vs TextCNN	805	1271	2076	104.1546	1.871e-24	Yes
Anshika Mean Pool vs BiGRU-Attn	648	1233	1881	181.3163	5.001e-41	Yes
Viraat RNN vs LSTM	601	822	1423	34.0126	1.095e-08	Yes
Viraat RNN vs TCN	530	734	1264	32.6021	1.131e-08	Yes

Robustness slices - Anshika (macro-F1 / error rate)

Slice	Mean Pool F1	Mean Pool err	TextCNN F1	TextCNN err	BiGRU F1	BiGRU err
Short Reviews	0.9224	7.50%	0.9342	6.35%	0.9385	5.93%
Medium Reviews	0.9316	6.84%	0.9453	5.47%	0.9474	5.26%
Long Reviews	0.9277	6.99%	0.9384	5.97%	0.9427	5.57%
Contains Negation	0.9219	7.55%	0.9384	5.97%	0.9420	5.63%
High Oov Rate	0.9138	8.22%	0.9244	7.19%	0.9253	7.11%

Robustness slices - Viraat (macro-F1 / error rate)

Slice	RNN F1	RNN err	LSTM F1	LSTM err	TCN F1	TCN err
Short Reviews	0.9420	5.59%	0.9425	5.57%	0.9435	5.48%
Medium Reviews	0.9431	5.69%	0.9518	4.82%	0.9484	5.16%
Long Reviews	0.9379	5.98%	0.9438	5.39%	0.9480	4.98%
Contains Negation	0.9379	6.02%	0.9462	5.19%	0.9449	5.31%
High Oov Rate	0.9225	7.38%	0.9273	6.96%	0.9265	7.05%

2.4.2 What we would try next

- Evaluate a scratch-trained chunked or hierarchical long-review model on a new validation/test protocol rather than retuning against the consumed official test set. The 256-token representation truncates 8,983/504,000 training reviews (1.782%), and the frozen qualitative review contains late-resolution errors where decisive evidence appears after long negative setup.
- Target discourse reversal, temporal updates and aspect conflict with an order-sensitive scratch encoder and predefined future slices. These error types recur in both members' 20-case reviews, while the high-OOV slice remains the weakest robustness slice for the strongest models (Anshika BiGRU-Attn macro-F1 0.9253; Viraat LSTM 0.9273).
- Compare a character/subword representation learned from scratch against the current word-level vocabulary while keeping the 0.50 threshold, training split and model-selection protocol frozen. This directly tests informal spelling/OOV failure modes without violating the no-pretrained-embedding/no-pretrained-LM constraint.

3. Task 3 - CycleGAN unpaired Monet/photo translation

Task 3 trains CycleGAN models for unpaired bidirectional translation between Monet paintings and photographs. Each system contains two generators and two discriminators and combines least-squares adversarial loss with cycle-consistency and identity losses. Evaluation uses frozen predictions in both directions plus the unchanged instructor FID/MiFID evaluator.

3.1 Anshika Goel - nine-block ResNet CycleGAN, selected epoch 125

Anshika's final selected configuration is epoch 125 of run `cyclegan_baseline_rtx4090_run001`. It uses two nine-block ResNet generators and two 70x70 PatchGAN discriminators. Instance normalization and tanh outputs are used in the generators, while least-squares GAN loss, cycle L1, identity L1, and a replay pool support unpaired training. The selected checkpoint followed deterministic comparison of later checkpoints with epoch 50 retained as a reproduction control.

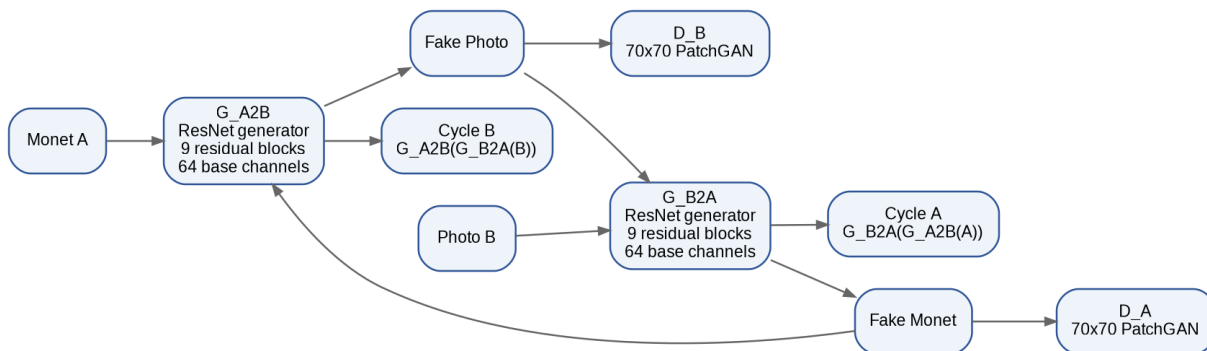


Figure 25. Anshika Task 3 CycleGAN architecture with nine residual blocks per generator and two PatchGAN discriminators.

Architecture and training configuration

Setting	Value
Generators	2 ResNet generators, 9 residual blocks each, 64 base channels
Discriminators	2 x 70x70 PatchGAN, 64 base channels
Normalization / output	InstanceNorm / tanh
Trainable parameters	28,285,832
Image transform	Resize 286 -> random crop 256; horizontal flip p=0.5
Batch / precision	1 / FP32
Optimizer	Adam, LR 0.0002, betas (0.5, 0.999), weight decay 0
Cycle L1 weight	10 per direction
Identity effective weight	5 per direction
Replay pool	50 images/domain
Configured schedule	200 epochs; 100 constant + 100 linear decay
Selected checkpoint	Epoch 125, global step 879,750
Final-epoch LR	0.00015
Training hardware	NVIDIA GeForce RTX 4090

Final automatic metrics and leaderboard evidence

Metric	Monet -> Photo	Photo -> Monet	Mean / official
FID	96.068450	98.864013	Official avg 97.466148498
MiFID	0.413548	0.400183	Official avg 0.406865570
KID	0.017190	0.007900	0.012545
Generative precision	0.7800	0.4333	0.6067
Generative recall	0.3333	0.6300	0.4817
Cycle L1	0.032256	0.034622	0.033439
Cycle LPIPS (AlexNet)	0.182873	0.134368	0.158620
Content cosine	0.783451	0.754759	0.769105

Training/resource metric	Value
Completed epochs / final step	125 / 879,750
Training duration	27.8685 hours
Weighted throughput	17.5377 domain images/s
Nonfinite gradient events	0
Final generator gradient norm	34.309808
Final discriminator A / B gradient norms	4.317097 / 6.441369
Final cycle loss A / B	0.543312 / 0.627509

Training/resource metric	Value
Final identity loss A / B	0.147671 / 0.203830
Final discriminator loss A / B	0.055120 / 0.157722
Final total generator loss	2.811135
Kaggle score	-48.9365
Final selected checkpoint SHA256	3216b4997ed51962590251170ca5cbd6d00fb444479a8b5cd9480f0cfbeab7ec
Peak CUDA memory for final epoch-125 continuation	Not retained in final training_run_summary (historical epoch-50 run recorded approx. 2542 MiB)

Interpretation: Epoch 125 improves the instructor FID/MiFID and Kaggle score relative to the earlier epoch-75 evaluation. The final model has strong cycle-reconstruction metrics but direction-specific tradeoffs: Monet->Photo has the better FID/KID while Photo->Monet has better MiFID and recall. Content cosine is a proxy, not a complete semantic-content measure.

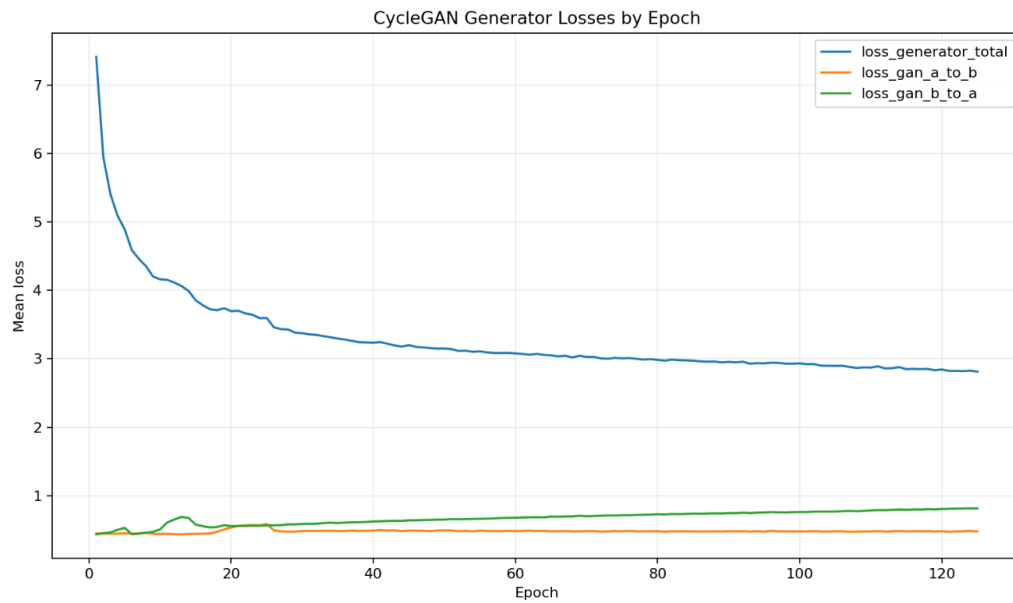


Figure 26. Generator losses for Anshika Task 3.

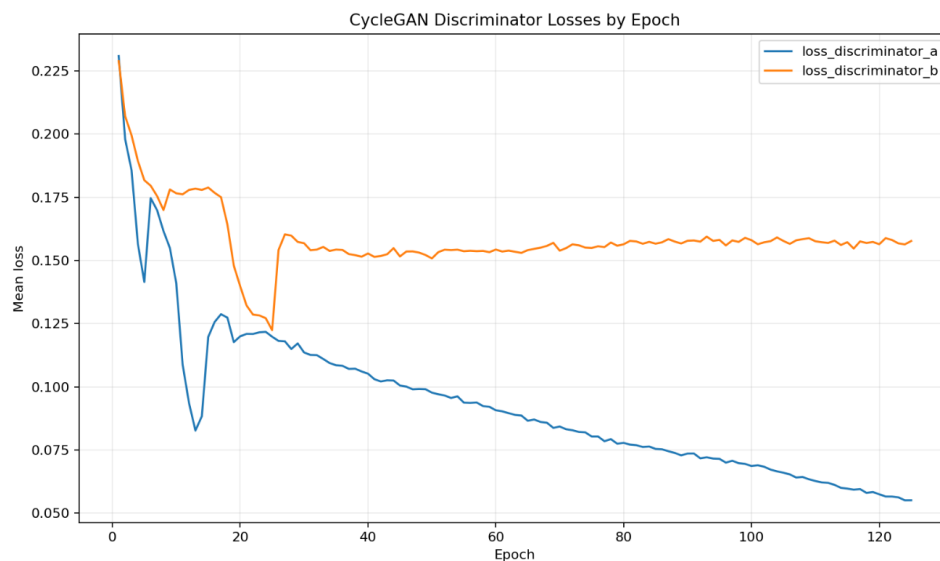


Figure 27. Discriminator losses for Anshika Task 3.

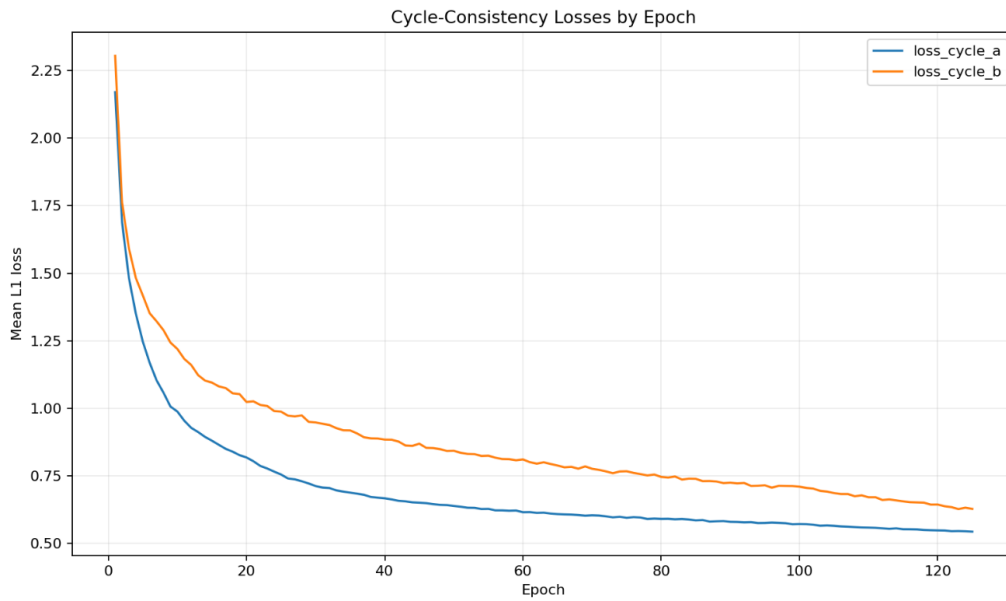


Figure 28. Cycle-consistency losses for Anshika Task 3.

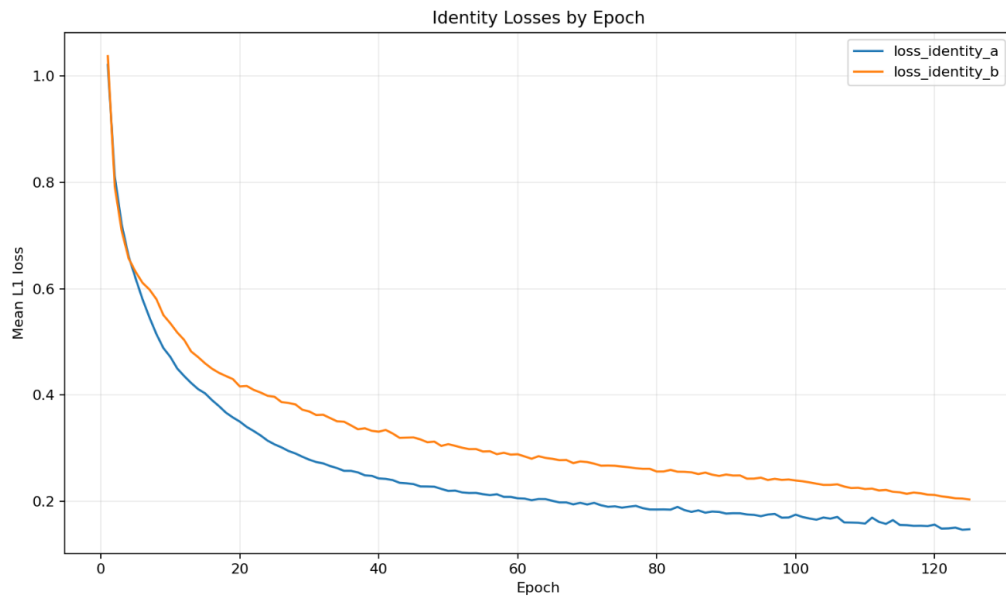


Figure 29. Identity losses for Anshika Task 3.

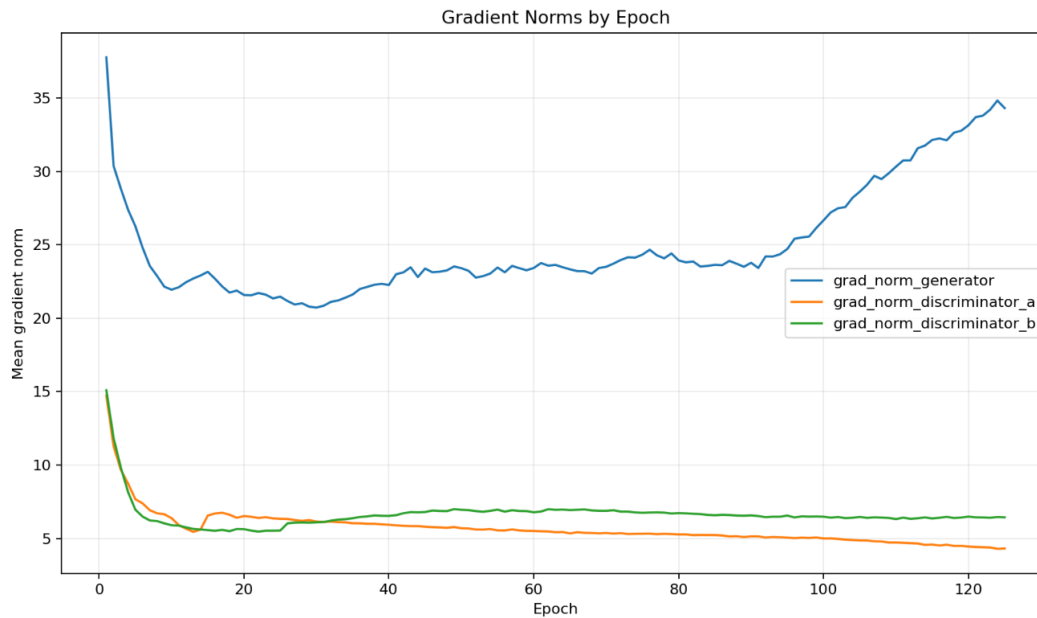


Figure 30. Gradient norms for Anshika Task 3.

```

print("\n Evaluating Photo -> Monet (B2A) ")
# Ground truth = real Monet; Generated = pred_B2A (generated Monet-like)
fid_B2A, mifid_B2A = calculate_fid_mifid(real_monet, gen_b2a, batch_size=BATCH_SIZE)
print(f"[Photo->Monet] FID={fid_B2A:.3f} MiFID={mifid_B2A:.4f}")

print("\n Evaluating Monet -> Photo (A2B) ")
# Ground truth = real Photo; Generated = pred_A2B (generated Photo-like)
fid_A2B, mifid_A2B = calculate_fid_mifid(real_photo, gen_a2b, batch_size=BATCH_SIZE)
print(f"[Monet->Photo] FID={fid_A2B:.3f} MiFID={mifid_A2B:.4f}")

Evaluating Photo -> Monet (B2A)
Inception activations: 100%|██████████| 10/10 [00:01<00:00, 5.11it/s]
Inception activations: 100%|██████████| 10/10 [00:02<00:00, 4.83it/s]
[Photo->Monet] FID=98.864 MiFID=0.4002

Evaluating Monet -> Photo (A2B)
Inception activations: 100%|██████████| 10/10 [00:01<00:00, 5.53it/s]
Inception activations: 100%|██████████| 10/10 [00:02<00:00, 4.93it/s]
[Monet->Photo] FID=96.068 MiFID=0.4135

import pandas as pd

sub_fid = (fid_A2B + fid_B2A) / 2
sub_mifid = (mifid_A2B + mifid_B2A) / 2

submission = pd.DataFrame([
    {"ID": 1,
     "FID": float(sub_fid),
     "MiFID": float(sub_mifid)}])

submission.to_csv("submission.csv", index=False)
print(submission)

```

ID	FID	MiFID	
0	1	97.466148	0.406866

Figure 31. Executed instructor evaluator evidence

```

() inference_manifest.json x
task3_gan > Anshika_Goel > outputs > cyclegan_baseline_rx4090_run001 > official_epoch125_predictions > () inference_manifest.json > ...
1  {
2    "final_model": true,
3    "checkpoint": {
4      "epoch": 125,
5      "global_step": 879750,
6      "sha256": "3216b4997ed51962590251170ca5cbd6d00fb444479a8b5cd9480f0cfbeab7ec"
7    },
8    "source": "outputs/improvement_checkpoint_comparison_epoch80_to125/epoch_0125",
9    "pred_A2B_count": 300,
10   "pred_B2A_count": 300,
11   "pred_A2B_aggregate_sha256": "6adf29c63c2a96df7675dd3b0eb1fba462e18cde5b5df7532596494fa1d457ec",
12   "pred_B2A_aggregate_sha256": "ab389066a9dd6b03e78c9a78529abf8ab48a33579e8b52359e35297c53a21d41",
13   "submitted_to_kaggle": true,
14   "kaggle_score": -48.9365,
15   "kaggle_rank_snapshot": 14
16 }
17

```

Figure 32. Final prediction provenance

Evaluation analysis - grounded evidence

Anshika's selected epoch-125 CycleGAN is supported by three independent evidence layers: the unchanged instructor evaluator, the deterministic directional metric pipeline, and the preserved training/inference provenance. The unchanged instructor notebook reports an official average FID of 97.466148498 and MiFID of 0.406865570. On the same frozen 300-image subset per direction, the directional pipeline gives Monet -> Photo FID 96.068450, MiFID 0.413548, KID 0.017190, precision 0.7800, recall 0.3333; and Photo -> Monet FID 98.864013, MiFID 0.400183, KID 0.007900, precision 0.4333, recall 0.6300. The model is therefore asymmetric by direction: Monet -> Photo is stronger on FID and precision, while Photo -> Monet is stronger on MiFID/KID and recall.

The reconstruction metrics provide a second view of quality. Cycle L1 is 0.032256 for A2B2A and 0.034622 for B2A2B (mean 0.033439), while AlexNet LPIPS is 0.182873 and 0.134368 (mean 0.158620). These values indicate that the round-trip mappings reconstruct the source relatively closely in pixel and perceptual feature space. Direct source-to-translation content cosine, however, is lower: 0.783451 for Monet -> Photo and 0.754759 for Photo -> Monet, with a mean of 0.769105. Because cycle consistency is a round-trip measure while content cosine compares the source with the direct translation, the two together show why low reconstruction error cannot by itself prove that all source semantics are preserved in the intermediate translated image.

From epoch 75 to epoch 125, mean FID improved from 98.170107 to 97.466231 and mean MiFID from 0.410627 to 0.406866; mean cycle L1 also improved from 0.038552 to 0.033439. At the same time, KID moved from 0.011541 to 0.012545, recall from 0.5000 to 0.4817, and content cosine from 0.795510 to 0.769105. Precision changed only slightly, from 0.6050 to 0.6067. Thus epoch 125 is the best retained checkpoint, but it is not uniformly superior on every auxiliary metric.

Metric	Epoch 75	Epoch 125	Observed change
FID mean	98.170107	97.466231	Improved
MiFID mean	0.410627	0.406866	Improved
KID mean	0.011541	0.012545	Worsened
Precision mean	0.6050	0.6067	Slightly improved
Recall mean	0.5000	0.4817	Worsened
Content cosine	0.795510	0.769105	Worsened
Cycle L1	0.038552	0.033439	Improved

The durable event record also supports training stability rather than merely final image quality. The selected run completed 125 epochs and global step 879,750, recorded 27.8685 training hours at a weighted throughput of 17.5377

domain images/s, and logged zero non-finite gradient events. The final prediction bundle contains exactly 300 Monet -> Photo and 300 Photo -> Monet translations tied to the selected epoch-125 checkpoint through its inference manifest.

Failure analysis - final epoch-125 model

The final epoch-125 failure analysis is primarily a trade-off analysis rather than a numerical-instability diagnosis: the run has zero logged non-finite gradient events, yet the auxiliary metrics show that stronger target-domain alignment was obtained with weaker source/translation feature similarity. Mean content cosine at epoch 125 is 0.769105, compared with the epoch-50 reproduction-control value of 0.806853. This does not prove semantic content loss for every image, but it is direct evidence that continued style/adversarial optimization changed the feature-space relationship between source and direct translation even while the official FID/MiFID improved.

A second failure mode is direction-dependent coverage. Monet -> Photo has high generative precision (0.7800) but low recall (0.3333), indicating that its generated outputs are concentrated in regions judged to resemble the target photo distribution but cover less of that distribution. Photo -> Monet shows the opposite tendency: lower precision (0.4333) but higher recall (0.6300). The mean precision/recall pair of 0.6067/0.4817 therefore hides substantial directional asymmetry.

A third limitation is metric disagreement. Relative to epoch 75, epoch 125 improves FID, MiFID, and cycle L1, while KID becomes slightly worse, recall declines, and content cosine declines. The selected checkpoint should therefore be described as better under the official evaluator and competition criterion, not as universally better on every property of the translations. Cycle L1/LPIPS quantify round-trip reconstruction, whereas FID/MiFID/KID quantify distributional behavior and content cosine is only a feature-space proxy; no one metric is sufficient to characterize translation quality.

Epoch 75 ? Epoch 125 auxiliary comparison		
Metric	Epoch 75	Epoch 125
FID mean	98.170107	97.466231
MiFID mean	0.410627	0.406866
KID mean	0.011541	0.012545
Precision mean	0.6050	0.6067
Recall mean	0.5000	0.4817
Content cosine	0.795510	0.769105
Cycle L1	0.038552	0.033439

Figure 33. Anshika Task 3 final-model failure/limitation evidence.

Strengths, weaknesses, limitations, and design rationale

- Strengths: best team official average FID and MiFID; better cycle L1/LPIPS reconstruction; improved Kaggle score at epoch 125; stable long training with no nonfinite-gradient events.
- Weaknesses: larger model and much longer training than Viraat; lower content cosine; direction-specific recall/precision tradeoffs; final continuation summary does not preserve a peak-memory value.
- Limitations: metrics are based on 300 fixed images per direction; FID/KID/LPIPS are estimator/proxy metrics; historical simulated ratings are retained only as provenance; the required blinded two-rater audit of the final models is reported in Section 3.4.
- Why this design: nine ResNet blocks follow the high-capacity CycleGAN pattern for 256x256 translation, cycle weight 10 constrains content consistency, and identity weight 5 discourages unnecessary changes when an image is already in the target domain.

3.2 Viraat Chaudhary - six-block resize-convolution CycleGAN

Viraat adapted the shared CycleGAN training infrastructure but changed the generator design to six residual blocks and nearest-neighbor resize followed by reflection-padded convolution. This reduces capacity and changes the upsampling

operator. The final run completed all 60 configured epochs with BF16 autocast while retaining FP32 model and optimizer state.

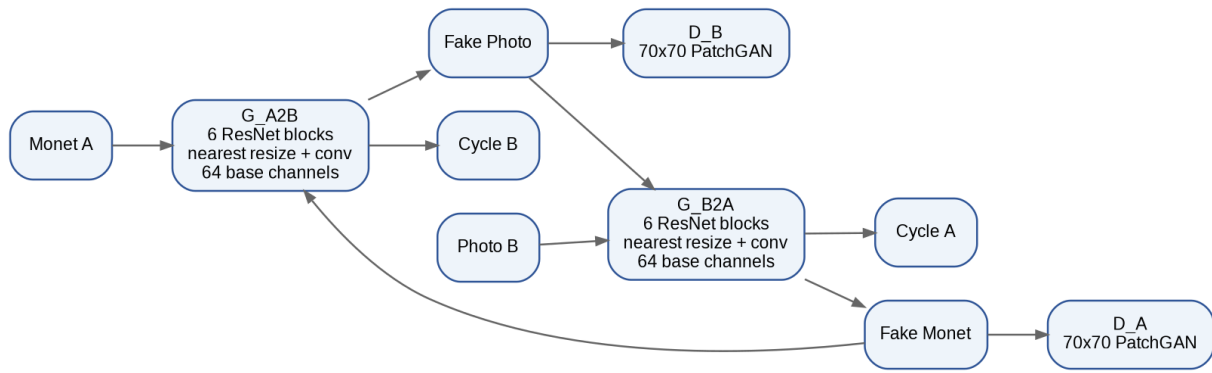


Figure 34. Viraat Task 3 CycleGAN architecture: six-block ResNet generators with nearest-resize convolution upsampling.

Architecture and training configuration

Setting	Value
Generators	2 RGB ResNet generators, 6 residual blocks, 64 base channels, nearest resize + reflection-padded convolution
Discriminators	2 three-layer 70x70 PatchGANs, 64 base channels
Trainable parameters	21,204,872
Training domains	300 Monet / 7,038 photos, unpaired
Seed	2661510
Transform	Resize 286; random crop 256; flip p=0.5; normalize to [-1,1]
Batch / precision	2 / BF16 autocast; FP32 model and Adam states
Optimizer	Adam; LR 0.0002; betas (0.5,0.999)
Schedule	20 constant epochs + 40 linear-decay epochs
Cycle weight	10
Identity ratio / effective weight	0.25 / 2.5
Replay pool	50
Final epoch	60; global step 211,140
Hardware	NVIDIA GeForce RTX 4090

Final automatic metrics

Metric	Monet -> Photo	Photo -> Monet	Mean / official
FID	103.036489	95.615238	Official avg 99.328525404
MiFID	0.416142	0.405585	Official avg 0.410869756
KID	0.024615	0.007861	-
Generative precision	0.623333	0.546667	-
Generative recall	0.426667	0.596667	-
Cycle L1	0.043438	0.046902	0.045170

Metric	Monet -> Photo	Photo -> Monet	Mean / official
Cycle LPIPS	0.420099	0.309908	0.365004
Content cosine	0.789136	0.793367	0.791252

Training/resource metric	Value
Completed epochs / steps	60 / 211,140
Recorded training time	18,960.139 s (5.2667 h)
Throughput	44.544 domain images/s
Peak allocated CUDA memory	2,587.737 MiB
Nonfinite gradient count	0
Final generator grad norm	32.479166
Final discriminator A / B grad norm	4.104792 / 5.933052
Final cycle loss A / B	0.743626 / 0.822668
Final identity loss A / B	0.155582 / 0.202499
Final discriminator loss A / B	0.038158 / 0.042537
Production checkpoint SHA256	138f9ae9edef412125ddfad643d2fc672c3de2d609819caf0b623979d3765252

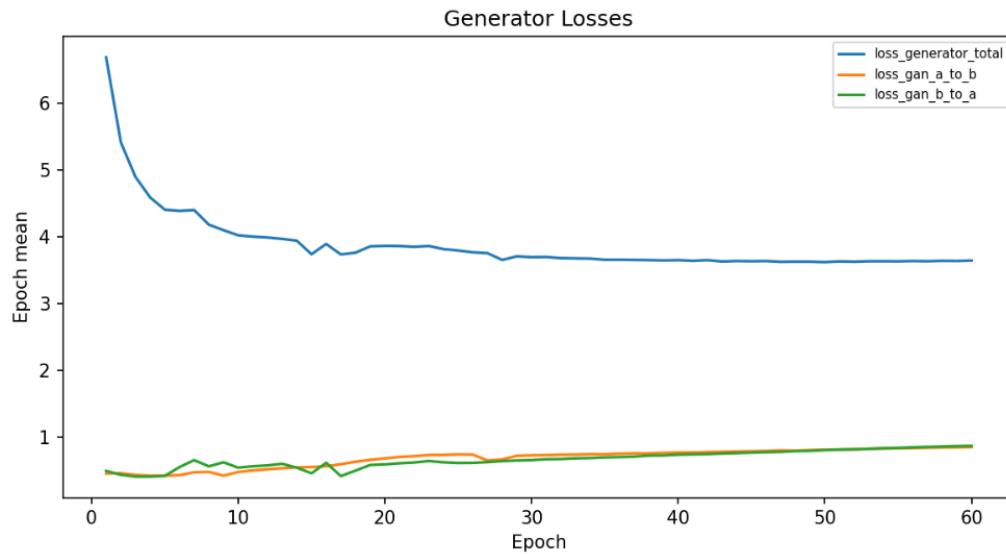


Figure 35. Generator losses for Viraat Task 3.

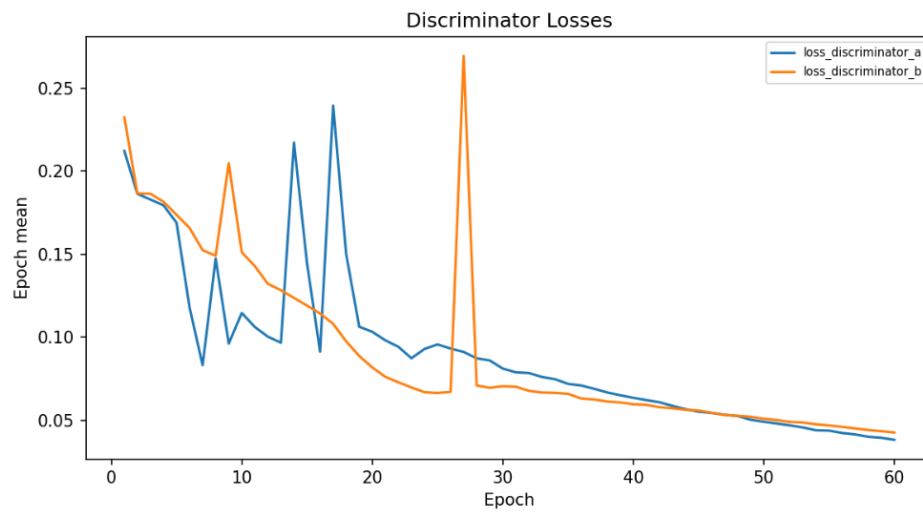


Figure 36. Discriminator losses for Viraat Task 3.

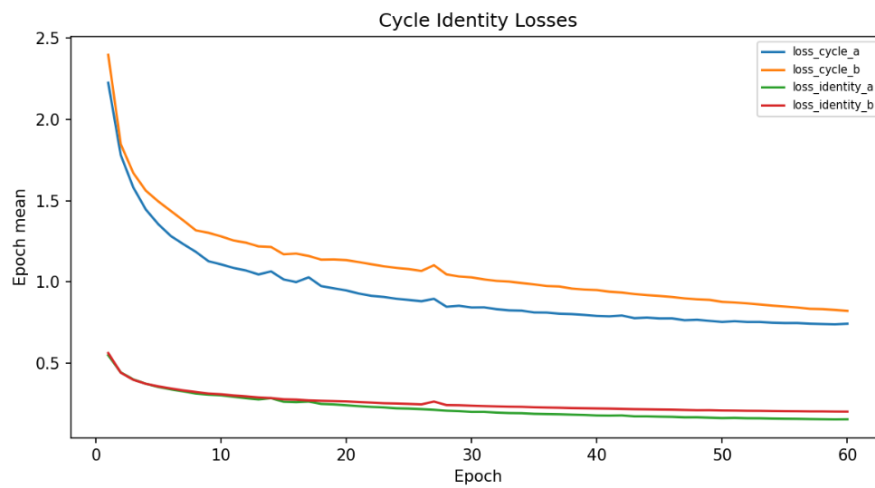


Figure 37. Cycle and identity losses for Viraat Task 3.

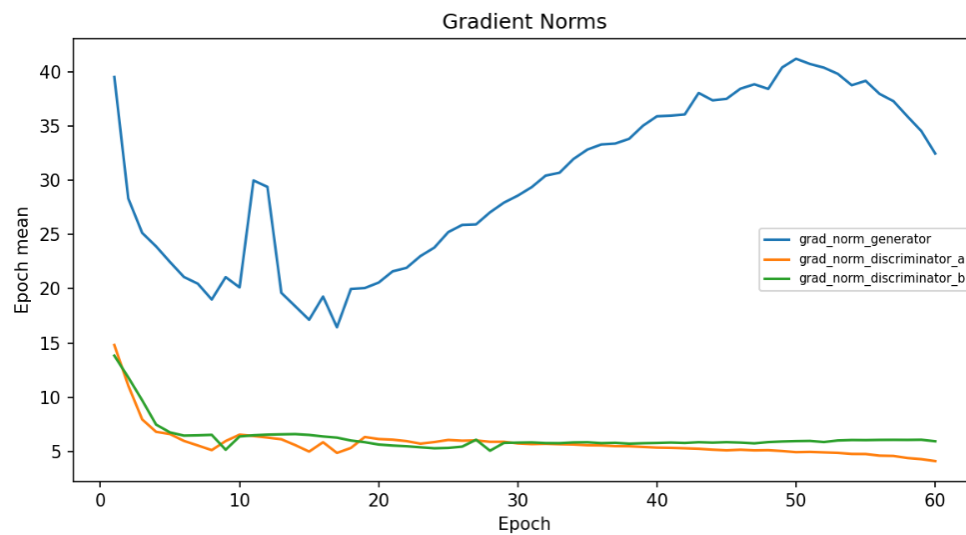


Figure 38. Gradient norms for Viraat Task 3.

```

1 (events) joni >
2 task_gan > viarai_chaudhary > logs > viarai_resizemc_run001 > (1) events.json
3
4 {"event_type": "epoch_end", "payload": {"batch_count": 14, "batch_index": 14, "batch_index": 12398, "epoch": 59, "epoch_seed": 2788093133643734680, "global_step": 1260500, "learning_rate": 0.75609756097562e-06, "metrics": {"fake_a_max": 0.0, "fake_a_mean": 0.0, "fake_b_max": 0.0, "fake_b_mean": 0.0, "real_a_max": 0.0, "real_a_mean": 0.0, "real_b_max": 0.0, "real_b_mean": 0.0, "real_c_max": 0.0, "real_c_mean": 0.0, "real_d_max": 0.0, "real_d_mean": 0.0, "real_e_max": 0.0, "real_e_mean": 0.0, "real_f_max": 0.0, "real_f_mean": 0.0, "real_g_max": 0.0, "real_g_mean": 0.0, "real_h_max": 0.0, "real_h_mean": 0.0, "real_i_max": 0.0, "real_i_mean": 0.0, "real_j_max": 0.0, "real_j_mean": 0.0, "real_k_max": 0.0, "real_k_mean": 0.0, "real_l_max": 0.0, "real_l_mean": 0.0, "real_m_max": 0.0, "real_m_mean": 0.0, "real_n_max": 0.0, "real_n_mean": 0.0, "real_o_max": 0.0, "real_o_mean": 0.0, "real_p_max": 0.0, "real_p_mean": 0.0, "real_q_max": 0.0, "real_q_mean": 0.0, "real_r_max": 0.0, "real_r_mean": 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LPIPS 0.420099/0.309908 for A2B2A/B2A2B. Thus, Viraat preserves source-to-translation feature similarity relatively well, especially in B2A, but its round-trip perceptual reconstruction is weaker.

Training evidence supports numerical stability and efficiency rather than a claim of universal visual superiority. The run completed all 60 configured epochs and global step 211,140 in 5.2667 hours at 44.544 domain images/s, with 2,587.737 MiB peak allocated CUDA memory and zero logged non-finite gradient events. Across training, cycle losses fell from 2.227211/2.399329 at epoch 1 to 0.743626/0.822668 at epoch 60; identity losses fell from 0.549718/0.563311 to 0.155582/0.202499; and total generator loss fell from 6.683079 to 3.640399. These trends show that the optimization remained finite and that reconstruction/identity objectives improved substantially, while the distributional evaluation metrics still expose an A2B weakness.

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A2B	fid	103.036489	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
A2B	class_mifid	0.4161415	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
A2B	kid_mean	0.02461479	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
A2B	kid_std	0.00340651	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
A2B	generative_precision	0.62333333	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
A2B	generative_recall	0.42666667	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
A2B	content_preservation_cosine_similarity	0.78913648	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
B2A	fid	95.6152381	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
B2A	class_mifid	0.40558487	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
B2A	kid_mean	0.00786131	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
B2A	kid_std	0.00124407	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
B2A	generative_precision	0.54666667	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
B2A	generative_recall	0.59666667	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
B2A	content_preservation_cosine_similarity	0.79336668	computed	outputs/viraat_resizeconv6_run001/metrics/inception_features.npz
A2B	cycle_l1_rgb01	0.04343769	computed	outputs/viraat_resizeconv6_run001/metrics/cycle_metrics_per_sample.csv
A2B	cycle_lpips_alex	0.42009864	computed	outputs/viraat_resizeconv6_run001/metrics/cycle_metrics_per_sample.csv
B2A	cycle_l1_rgb01	0.04690231	computed	outputs/viraat_resizeconv6_run001/metrics/cycle_metrics_per_sample.csv
B2A	cycle_lpips_alex	0.30990798	computed	outputs/viraat_resizeconv6_run001/metrics/cycle_metrics_per_sample.csv
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both	training_seconds	18960.1394	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	images_per_second	44.5439764	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	peak_cuda_memory_mib	2587.7373	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	nonfinite_gradient_count	0	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	final_epoch_grad_norm_discriminator_a	4.10479241	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	final_epoch_grad_norm_discriminator_b	5.93305222	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	final_epoch_grad_norm_generator	32.4791663	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	final_epoch_loss_cycle_a	0.74362562	computed	logs/viraat_resizeconv6_run001/events.jsonl
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both	final_epoch_loss_gan_b_to_a	0.86792657	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	final_epoch_loss_generator_total	3.64039948	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	final_epoch_loss_identity_a	0.15558201	computed	logs/viraat_resizeconv6_run001/events.jsonl
both	final_epoch_loss_identity_b	0.20249923	computed	logs/viraat_resizeconv6_run001/events.jsonl

Figure 41. Grounded automatic-evaluation evidence for Viraat's completed epoch-60 CycleGAN.

Failure analysis - six fixed high-LPIPS cases

Viraat's failure analysis uses a fixed, reproducible selection rule rather than hand-picking visually poor images. The saved per-sample cycle CSV was used to identify the three highest cycle-LPIPS cases in each translation direction, after

which those six fixed panels were visually reviewed. The six retained IDs and round-trip reconstruction metrics are listed below.

Case	Panel path	Cycle L1	Cycle LPIPS
AUDIT_004	.../human_audit/AUDIT_004.png	0.034307	0.540362
AUDIT_001	.../human_audit/AUDIT_001.png	0.052304	0.519985
AUDIT_012	.../human_audit/AUDIT_012.png	0.055689	0.483064
AUDIT_002	.../human_audit/AUDIT_002.png	0.057657	0.463376
AUDIT_015	.../human_audit/AUDIT_015.png	0.071630	0.401893
AUDIT_023	.../human_audit/AUDIT_023.png	0.064907	0.394317

The highest cycle-LPIPS case, AUDIT_004, has cycle L1 0.034307 and LPIPS 0.540362. Its relatively modest L1 but very high perceptual distance demonstrates why pixel error and perceptual error cannot be treated as interchangeable measures. AUDIT_001 (L1 0.052304, LPIPS 0.519985) is explicitly documented as darkening with residual brush-like texture, showing exposure/color drift even when the broad scene remains recognizable.

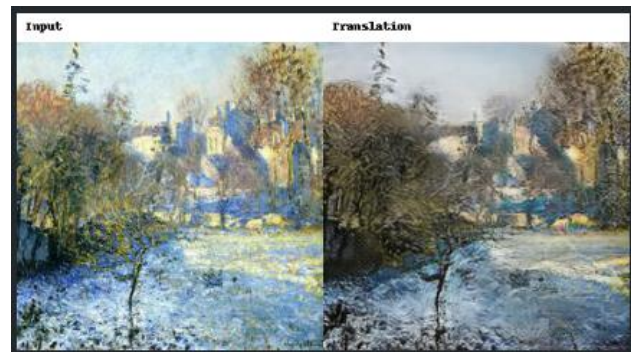
AUDIT_012 (L1 0.055689, LPIPS 0.483064) shows flowering vegetation becoming darker with dense repeated bright strokes; the sky shifts to a much darker gradient and fine flower/leaf detail is replaced by coarse texture, although the broad horizon remains. AUDIT_002 (L1 0.057657, LPIPS 0.463376) retains the lake/tree skyline and reflection layout but softens thin trunks and cloud boundaries and introduces strong blue-green color in some tree regions. These cases show a concrete style/content trade-off rather than complete scene destruction.

AUDIT_015 has the largest cycle L1 among the six selected cases (0.071630; LPIPS 0.401893), identifying it as the strongest pixel-space round-trip reconstruction failure in the fixed panel. AUDIT_023 (L1 0.064907, LPIPS 0.394317) preserves the approximate horizon and shoreline, but changes an orange/black sunset into a bright blue-green textured scene; small foreground figures/objects become less distinct and the sun/reflection contrast changes substantially. This is direct evidence that broad geometry can survive while local color, contrast, and fine content are altered.

The failure-analysis document also places an important boundary on interpretation: cycle metrics measure round-trip reconstruction, not direct translation correctness. A small cycle L1 does not establish good photographic or Monet style, and a large LPIPS value does not by itself identify where the direct translation failed. For follow-up work, the repository proposes controlled tests of six versus nine residual blocks, identity ratio 0.25 versus 0.5, and cycle weight 10 versus 12.5 while keeping the remaining configuration fixed; these are proposed experiments, not demonstrated improvements.



AUDIT_004.png

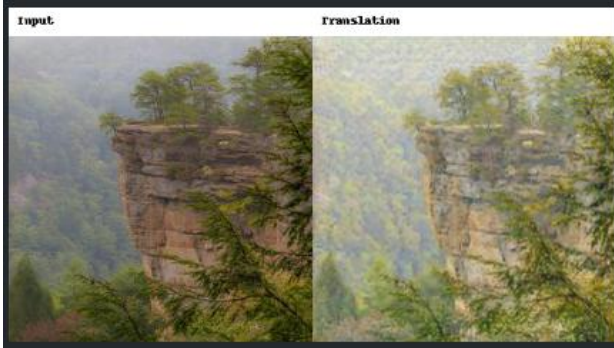


AUDIT_001.png



AUDIT_012.png

AUDIT_002.png



AUDIT_015.png



AUDIT_023.png

Figure 42. Viraat Task 3 fixed failure panel: three highest cycle-LPIPS cases per translation direction, selected from the saved per-sample metric CSV before visual interpretation.

Strengths, weaknesses, limitations, and design rationale

- Strengths: approximately 25% fewer parameters than Anshika; over 2.5x recorded throughput; much shorter total training; better Photo->Monet FID/KID; higher mean content cosine.
- Weaknesses: worse official average FID/MiFID and substantially worse cycle LPIPS; A2B FID is notably higher; lower capacity may limit fine reconstruction detail.
- Limitations: six-block generator, resize-convolution, identity weight, batch size/precision, and training schedule differ simultaneously; therefore the result does not isolate the effect of a single design choice.
- Why this design: six residual blocks and resize-convolution reduce compute/capacity and avoid relying on transposed-convolution upsampling, while BF16 and batch 2 prioritize throughput on the same RTX 4090 class of serious training hardware.

3.3 Task 3 team comparison

Property / metric	Anshika epoch 125	Viraat epoch 60	Observed advantage
Residual blocks	9	6	Viraat efficiency
Parameters	28.286 M	21.205 M	Viraat
Training time	27.868 h	5.267 h	Viraat
Throughput	17.538 img/s	44.544 img/s	Viraat
Official average FID	97.466148	99.328525	Anshika
Official average MiFID	0.406866	0.410870	Anshika
Monet -> Photo FID	96.068450	103.036489	Anshika
Photo -> Monet FID	98.864013	95.615238	Viraat
Monet -> Photo KID	0.017190	0.024615	Anshika
Photo -> Monet KID	0.007900	0.007861	Near tie / Viraat

Property / metric	Anshika epoch 125	Viraat epoch 60	Observed advantage
Mean cycle L1	0.033439	0.045170	Anshika
Mean cycle LPIPS	0.158620	0.365004	Anshika
Mean content cosine	0.769105	0.791252	Viraat

Interpretation: If the primary criterion is the course official evaluator plus cycle-reconstruction quality, Anshika's epoch-125 model is the stronger final configuration. Viraat offers a compelling efficiency/content-preservation tradeoff: far lower training cost, fewer parameters, better Photo->Monet FID, and higher content cosine. Neither model dominates every direction or every metric.

Architecture and hyperparameters		
Setting	Anshika (reported)	Viraat (saved run)
Generator	9 residual blocks, 64 base channels	6 residual blocks, 64 base channels; nearest resize-convolution
Discriminators	Two 70×70 PatchGANs	Two 70×70 PatchGANs
Parameters	28,285,832	21,204,872
Selected/evaluated epoch	125; configuration plans 200	60 of 60
Batch size / precision	1 / FP32	2 / BF16 autocast, FP32 weights/Adam
Configured LR schedule	100 constant epochs; 100 decay epochs	20 constant epochs; 40 decay epochs
Base LR / Adam betas	0.0002 / (0.5, 0.999)	0.0002 / (0.5, 0.999)
Identity ratio / effective weight	0.5 / 5	0.25 / 2.5
Cycle weight	10 each direction	10 each direction
Replay pool	50	50
Seed	20260929	2661510
Training GPU	Configuration names lab RTX 4090	RTX 4090 recorded in run manifest

Both-direction automatic metrics

A2B is Monet → Photo; B2A is Photo → Monet. Cycle measurements compare each source with its round trip. MiFID is the class evaluator's fixed-index feature cosine distance, not conventional memorization-informed FID.

Direction	Metric	Anshika epoch 125 (reported)	Viraat epoch 60 (local)
A2B	FID	96.068450	103.036489
A2B	MiFID	0.413548	0.416142
A2B	KID	0.017190	0.024615
A2B	Generative precision	0.780000	0.623333
A2B	Generative recall	0.333333	0.426667
A2B	Cycle reconstruction L1	0.032256	0.043438
A2B	Cycle LPIPS AlexNet	0.182873	0.420099
A2B	Inception cosine similarity	0.783451	0.789136
B2A	FID	98.864013	95.615238
B2A	MiFID	0.400183	0.405585
B2A	KID	0.007900	0.007861
B2A	Generative precision	0.433333	0.546667
B2A	Generative recall	0.630000	0.596667
B2A	Cycle reconstruction L1	0.034622	0.046902
B2A	Cycle LPIPS AlexNet	0.134368	0.309908
B2A	Inception cosine similarity	0.754759	0.793367

Evaluator summary	Anshika epoch 125 (reported)	Viraat epoch 60
Official professor FID average	97.466148498151	99.328525404097
Official professor MiFID average	0.406865569646	0.410869756014
Mean of local directional FIDs	97.466231244915	99.325863503503

Efficiency and final training state

Metric	Anshika epoch 125 (reported)	Viraat epoch 60
Recorded epoch-duration sum (hours)	27.868480	5.266705
Recorded domain images/sec	17.537734	44.543976
Peak allocated CUDA memory (MiB)	Not supplied for epoch 125	2587.737305
Logged nonfinite gradient count	0.000000	0.000000
loss_generator_total	2.811135	3.640399
loss_gan_a_to_b	0.475969	0.848098
loss_gan_b_to_a	0.812843	0.867927
loss_discriminator_a	0.055120	0.038158
loss_discriminator_b	0.157722	0.042537
loss_cycle_a	0.543312	0.743626
loss_cycle_b	0.627509	0.822668
loss_identity_a	0.147671	0.155582
loss_identity_b	0.203830	0.202499
grad_norm_generator	34.309808	32.479166
grad_norm_discriminator_a	4.317097	4.104792
grad_norm_discriminator_b	6.441369	5.933052

Figure 43. Task 3 side-by-side comparison.

3.3.1 Complete Task 3 team comparison matrix

This matrix consolidates the final selected CycleGAN configurations and every required Task 3 metric. Direction labels follow the report convention Monet -> Photo and Photo -> Monet.

Property / metric	Anshika epoch 125	Viraat epoch 60	Comparison
Generator architecture	2 x 9-block ResNet generators, 64 base channels	2 x 6-block ResNet generators, 64 base channels; nearest-resize + reflection-padded convolution	Capacity / upsampling differ
Discriminators	2 x 70x70 PatchGAN, 64 base channels	2 x three-layer 70x70 PatchGAN, 64 base channels	Both PatchGAN
Parameters	28,285,832	21,204,872	Viraat smaller
Batch / precision	1 / FP32	2 / BF16 autocast; FP32 states	Viraat efficiency
Optimizer	Adam; LR 0.0002; betas (0.5,0.999); wd 0	Adam; LR 0.0002; betas (0.5,0.999)	Matched LR/betas
Schedule	200 configured: 100 constant + 100 decay; selected epoch 125	60: 20 constant + 40 decay	Different budgets
Cycle weight	10 per direction	10	Matched
Identity effective weight	5 per direction	2.5	Anshika stronger identity penalty
Replay pool	50/domain	50	Matched
Training hardware	NVIDIA GeForce RTX 4090	NVIDIA GeForce RTX 4090	Same GPU class
FID Monet -> Photo	96.068450	103.036489	Anshika lower
FID Photo -> Monet	98.864013	95.615238	Viraat lower
Official average FID	97.466148498	99.328525404	Anshika lower
MiFID Monet -> Photo	0.413548	0.416142	Anshika lower
MiFID Photo -> Monet	0.400183	0.405585	Anshika lower
Official average MiFID	0.406865570	0.410869756	Anshika lower
KID Monet -> Photo	0.017190	0.024615	Anshika lower
KID Photo -> Monet	0.007900	0.007861	Near tie; Viraat lower
Generative precision Monet -> Photo	0.7800	0.623333	Anshika
Generative precision Photo -> Monet	0.4333	0.546667	Viraat
Generative recall Monet -> Photo	0.3333	0.426667	Viraat
Generative recall Photo -> Monet	0.6300	0.596667	Anshika
Cycle L1 Monet -> Photo	0.032256	0.043438	Anshika lower
Cycle L1 Photo -> Monet	0.034622	0.046902	Anshika lower
Cycle LPIPS Monet -> Photo	0.182873	0.420099	Anshika lower
Cycle LPIPS Photo -> Monet	0.134368	0.309908	Anshika lower

Property / metric	Anshika epoch 125	Viraat epoch 60	Comparison
Content cosine Monet -> Photo	0.783451	0.789136	Viraat higher
Content cosine Photo -> Monet	0.754759	0.793367	Viraat higher
Final cycle loss A / B	0.543312 / 0.627509	0.743626 / 0.822668	Anshika lower
Final identity loss A / B	0.147671 / 0.203830	0.155582 / 0.202499	Mixed / close
Final discriminator loss A / B	0.055120 / 0.157722	0.038158 / 0.042537	Different adversarial balance
Final generator grad norm	34.309808	32.479166	Both finite
Final discriminator grad norm A / B	4.317097 / 6.441369	4.104792 / 5.933052	Both finite
Nonfinite gradient events	0	0	Stable
Training time	27.8685 h	5.2667 h	Viraat
Throughput	17.5377 domain img/s	44.544 domain img/s	Viraat
Human style mean	3.2000	3.2500	Viraat +0.0500
Human content-preservation mean	3.4833	3.7500	Viraat +0.2667
Human artifact-free mean	3.4500	3.6833	Viraat +0.2333
Pooled exact agreement	27.78%	40.00%	Viraat raters agreed more often
Pooled Cohen's kappa	0.0782	0.2357	Both modest; Viraat higher

3.3.2 What we would try next

- Run a matched-budget multi-seed ablation that changes one factor at a time: six versus nine residual blocks, identity effective weight 2.5 versus 5, and batch/precision settings, while holding the image transform, optimizer, cycle weight, replay pool and training schedule fixed. The current comparison cannot isolate one architectural cause because these factors and total training duration differ together.
- Target the Monet -> Photo direction specifically. Anshika reaches high precision 0.7800 but low recall 0.3333, while Viraat has FID 103.036489 in the same direction; the next model should test whether coverage can be improved without sacrificing Anshika's lower FID/cycle error or Viraat's throughput advantage.
- Repeat the blinded visual evaluation with a larger fixed panel and additional independent raters while preserving the same anonymous protocol. The current pooled kappa is 0.0782 for Anshika and 0.2357 for Viraat, so a larger audit would quantify whether the small mean-rating advantage for Viraat is stable despite modest inter-rater agreement.

3.4 Blinded real-human audit

To complement the automatic Task 3 metrics, the final selected models were evaluated in a blinded real-human audit. For each model, 30 fixed panels were rated: 15 Monet -> Photo and 15 Photo -> Monet. The same two independent human raters scored every panel on three integer 1-5 scales: style quality, content preservation, and artifact-free quality; higher scores indicate better perceived quality. The raters received anonymous Set_A / Set_B packages, and model identity was revealed only after both completed responses were returned and validated. Historical simulated/AI-assisted ratings were not used in these final real-human calculations.

Overall human-rating results

Criterion	Anshika epoch 125	Viraat epoch 60	Viraat - Anshika
Style quality mean	3.2000	3.2500	+0.0500
Content preservation mean	3.4833	3.7500	+0.2667
Artifact-free quality mean	3.4500	3.6833	+0.2333

Interpretation: Viraat receives the higher two-rater mean on all three perceptual criteria. The style difference is only 0.05 on a five-point scale, so style quality is effectively very close. The larger gaps are content preservation (+0.2667) and artifact-free quality (+0.2333), which descriptively favor Viraat. These differences are descriptive human-audit results, not inferential significance tests: no p-value or confidence interval was computed for the human ratings, so they should not be presented as statistically significant differences.

Inter-rater agreement

Agreement metric	Anshika epoch 125	Viraat epoch 60	Meaning
Style exact agreement	16.67%	40.00%	Identical 1-5 score
Style Cohen's kappa	-0.0190	0.2275	Chance-adjusted agreement
Content exact agreement	26.67%	43.33%	Identical 1-5 score
Content Cohen's kappa	0.0517	0.2630	Chance-adjusted agreement
Artifact exact agreement	40.00%	36.67%	Identical 1-5 score

Agreement metric	Anshika epoch 125	Viraat epoch 60	Meaning
Artifact Cohen's kappa	0.2405	0.1775	Chance-adjusted agreement
Pooled exact agreement	27.78%	40.00%	All three criteria pooled
Pooled Cohen's kappa	0.0782	0.2357	All three criteria pooled

Significance of agreement metrics: Exact agreement is the percentage of samples for which the two raters chose the same integer score. Cohen's kappa adjusts that agreement for chance. Anshika's pooled kappa of 0.0782 shows very limited beyond-chance agreement overall, with style particularly subjective (kappa -0.0190). Viraat's pooled kappa is higher at 0.2357, but agreement is still modest rather than strong. Therefore, the human audit is useful as perceptual evidence, but the low-to-moderate rater consistency means it should corroborate, not replace, the automatic metrics.

Direction-specific human ratings

Model / direction	Style	Content	Artifact-free
Anshika Monet -> Photo	2.6000	3.3667	3.0333
Anshika Photo -> Monet	3.8000	3.6000	3.8667
Viraat Monet -> Photo	2.8667	3.5000	3.5000
Viraat Photo -> Monet	3.6333	4.0000	3.8667

Directional significance: Both models were rated higher for Photo -> Monet than for Monet -> Photo on all three human criteria. For Anshika, the largest directional gaps are in style (3.8000 vs 2.6000) and artifact-free quality (3.8667 vs 3.0333). Viraat shows the same pattern, with Photo -> Monet reaching 4.0000 for content preservation. This human preference is consistent with Viraat's stronger Photo -> Monet FID/KID and, for Anshika, with the stronger Photo -> Monet MiFID/KID and recall reported earlier, even though not every automatic metric moves in the same direction.

Human audit versus automatic metrics

Combined interpretation: The human audit does not overturn the automatic comparison; instead, it exposes a complementary trade-off. Anshika remains stronger on the course official average FID/MiFID and on cycle-reconstruction quality, while Viraat receives slightly higher mean human ratings for style, content preservation, and artifact-free quality. This disagreement is important because FID/MiFID/KID measure distributional behavior, cycle L1/LPIPS measure round-trip reconstruction, content cosine is a feature-space proxy, and human raters judge visible style/content/artifacts directly. The combined evidence therefore supports the report's existing conclusion that neither model is a universal winner across all evaluation dimensions.

model	scope	criterion	n_samples	exact_agreement_percent	cohen_kappa
Anshika_Goel_epoch125	overall	style_quality	30	16.666667	-0.019022
Anshika_Goel_epoch125	overall	content_preservation	30	26.666667	0.051724
Anshika_Goel_epoch125	overall	artifact_free	30	40	0.240506
Anshika_Goel_epoch125	overall	pooled_all_dimensions	90	27.777778	0.078159
Anshika_Goel_epoch125	A2B_monet_to_photo	style_quality	15	6.666667	-0.09375
Anshika_Goel_epoch125	A2B_monet_to_photo	content_preservation	15	20	0.032258
Anshika_Goel_epoch125	A2B_monet_to_photo	artifact_free	15	20	0.037433
Anshika_Goel_epoch125	A2B_monet_to_photo	pooled_all_dimensions	45	15.555556	-0.016647
Anshika_Goel_epoch125	B2A_photo_to_monet	style_quality	15	26.666667	-0.024845
Anshika_Goel_epoch125	B2A_photo_to_monet	content_preservation	15	33.333333	0.142857
Anshika_Goel_epoch125	B2A_photo_to_monet	artifact_free	15	60	0.454545
Anshika_Goel_epoch125	B2A_photo_to_monet	pooled_all_dimensions	45	40	0.171214
Viraat_Chauthary_final_run	overall	style_quality	30	40	0.227468
Viraat_Chauthary_final_run	overall	content_preservation	30	43.333333	0.263006
Viraat_Chauthary_final_run	overall	artifact_free	30	36.666667	0.177489
Viraat_Chauthary_final_run	overall	pooled_all_dimensions	90	40	0.235729
Viraat_Chauthary_final_run	A2B_monet_to_photo	style_quality	15	40	0.15625
Viraat_Chauthary_final_run	A2B_monet_to_photo	content_preservation	15	33.333333	0.142857
Viraat_Chauthary_final_run	A2B_monet_to_photo	artifact_free	15	26.666667	0.073034
Viraat_Chauthary_final_run	A2B_monet_to_photo	pooled_all_dimensions	45	33.333333	0.145028
Viraat_Chauthary_final_run	B2A_photo_to_monet	style_quality	15	40	0.258242
Viraat_Chauthary_final_run	B2A_photo_to_monet	content_preservation	15	53.333333	0.339623
Viraat_Chauthary_final_run	B2A_photo_to_monet	artifact_free	15	46.666667	0.25
Viraat_Chauthary_final_run	B2A_photo_to_monet	pooled_all_dimensions	45	46.666667	0.288069

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Figure 44. Final blinded two-rater human-audit summary and inter-rater agreement evidence.

4. Overall conclusions and remaining submission checks

Across the three tasks, the strongest model depends on the evaluation objective rather than a universal architecture rule. Anshika's Task 1 GPT is the clear quality winner under the common held-out evaluation, while Viraat's Task 1 model is substantially more efficient. In Task 2, Viraat's LSTM has the highest observed accuracy/macro-F1/MCC across the six-model table, with the TCN almost tied and Anshika's BiGRU-Attention the strongest of her three. In Task 3, Anshika's epoch-125 CycleGAN wins on the official FID/MiFID averages and reconstruction metrics, whereas Viraat's six-block resize-convolution model provides much better training efficiency and stronger Photo->Monet/content-cosine results.

5. References

1. Vaswani, A., et al. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems* 30.
2. Eldan, R., & Li, Y. (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? *arXiv:2305.07759*.
3. Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level Convolutional Networks for Text Classification. *Advances in Neural Information Processing Systems* 28. Yelp Review Polarity dataset.
4. Bai, S., Kolter, J. Z., & Koltun, V. (2018). An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. *arXiv:1803.01271*.

5. Zhu, J.-Y., Park, T., Isola, P., & Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. ICCV.
6. Heusel, M., et al. (2017). GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. NeurIPS. (FID).
7. Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2018). The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. CVPR. (LPIPS).